Highlights

Sensor noise, array design and helmet fit in on-scalp and cryogenic magnetoencephalography: a simulation study in adult and pediatric head models

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- With modeled brain noise, a dense OPM array has higher detectability than SQUID MEG
- The dense-array advantage is as large as the model's error in SQUID detectability
- Which system has the higher SNR depends on OPM noise, array design and SNR measure
- In scaled and template heads, the extra OPM advantage largely reflects helmet fit
- Idealized detectors detect weaker simulated spikes with dense OPMs than with SQUIDs

Sensor noise, array design and helmet fit in on-scalp and cryogenic magnetoencephalography: a simulation study in adult and pediatric head models

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Abstract

Optically pumped magnetometers (OPMs) can be placed on the scalp, where the magnetic field of cortical sources is stronger than at the superconducting quantum interference devices (SQUIDs) of conventional magnetoencephalography (MEG). For interictal epileptiform discharges, however, OPMs have shown a higher signal-to-noise ratio (SNR) than SQUIDs in some comparisons but not in others. A spherical-head analysis using point sensors at fixed standoffs and an assumed OPM-to-SQUID noise ratio predicted the largest on-scalp benefit for superficial sources and small heads. Using realistic head models, we examined how relative SNR depends on noise, array design and helmet fit. Under sensor, brain and environmental noise, we compared the matched-filter detectability of cortical sources between the 306-channel Neuromag SQUID system and two single-axis OPM arrays, a dense array (208 sensors) and a site-matched array at the SQUID sites (98 sensors). We placed an adult head, two downscaled copies and two infant templates each in the adult helmet and in a helmet fitted at the adult's median sensor-to-scalp distance. In the adult, brain noise dominated the noise of both systems and reversed the ranking obtained under sensor noise alone. At an OPM noise of 15 fT/√Hz, the dense array had higher detectability at almost all cortical sources (median ratio 1.14, 95 % parcel-bootstrap interval [1.12, 1.17]), whereas the site-matched array showed no clear difference. OPM noise and array design determined which system had higher detectability; the SQUID channels compared and the SNR measure determined which system a sensor-level SNR favored. However, the dense-array advantage was as large as the noise model's underestimate of SQUID detectability. With measured SQUID noise and modeled OPM noise, the ratio fell to 1.03 [0.99, 1.08]. In the adult helmet, the smaller heads had greater sensor-to-scalp distances and a larger dense-array advantage than the adult (by +0.44 to +1.07 dB). In the fitted helmet, with fewer dense-array sites on each smaller head, almost none of this excess remained (+0.02 to +0.05 dB). Under the modeled noise, the dense array detected simulated superficial spikes at a 22–32 % lower dipole strength than the SQUID system (adult, and smaller heads in the adult helmet), with idealized detectors at a nominal false-event rate. Overall, the results suggest that noise, array design and helmet fit, rather than sensor proximity alone, determine whether OPMs provide a higher SNR than SQUIDs, which is of interest when OPM and SQUID recordings are compared in pediatric and epilepsy studies.

Keywords: Magnetoencephalography (MEG), Optically pumped magnetometer (OPM), Superconducting quantum interference device (SQUID), Pediatric neuroimaging, Helmet fit, Noise covariance, Interictal spikes

1. Introduction

Conventional magnetoencephalography (MEG) uses superconducting quantum interference devices (SQUIDs) housed in a rigid cryogenic helmet. Optically pumped magnetometers (OPMs) measure the brain's magnetic field at room temperature and can therefore be worn on the scalp (Boto et al., 2017; Iivanainen et al., 2017). The SQUID helmet is built for adults, leaving a wide and uneven gap between its sensors and a child's head, whereas an on-scalp array can fit any head (Feys and De Tiège, 2024; Hill et al., 2019). In clinical epilepsy recordings, interictal discharges are larger with OPMs than with cryogenic MEG (Feys et al., 2022; Ren et al., 2025). The signal-to-noise ratio (SNR) of these discharges, however, was higher with OPMs in some comparisons (Feys et al., 2022, 2025; Ren et al., 2025) and equal or slightly lower in others (Hillebrand et al., 2023; Schwartz et al., 2025). The weakest OPM results concern temporal and mesial temporal sources (Feys et al., 2024, 2025; Shen et al., 2026).

One might expect on-scalp sensors to detect every source better because they are closer to it. However, an OPM is noisier than a SQUID magnetometer, so a larger field alone does not yield a higher SNR. A spherical-head analysis made this trade-off explicit, using a point magnetometer on the scalp for the OPM and one 18 mm above it for the SQUID (Jas et al., 2026). At intermediate OPM-to-SQUID noise ratios, the OPM had higher single-channel SNR only for sources shallower than the equal-SNR depth, the depth at which the two sensors have the same SNR. In an adult head, this depth was about 28 mm when OPM noise was three times SQUID noise. The SQUID sensors were placed at the same scalp distance for every head size, leaving a larger proportion of a small brain shallower than the equal-SNR depth. The analysis therefore predicts a larger OPM benefit in children.

A realistic comparison adds sensor, brain and room noise as each array records it, finite arrays on a real scalp, and a real SQUID helmet around a smaller head. Earlier simulations have included some of these elements. With sensor noise alone, a helium-4 OPM array improved relative to SQUID magnetometers as the head shrank, both in a standard adult placement and in a constant-gap helmet (Zahran et al., 2022). When brain noise was added in one adult, simulated spikes were visible on the scalp from smaller cortical patches with on-scalp than with conventional MEG (Westin et al., 2023). In contrast, with brain noise but no sensor noise modeled in epilepsy patients, OPMs at the SQUID sites did not raise channel-averaged SNR in any region. The OPM–SQUID difference nevertheless followed the ratio of sensor distances, which tracked head circumference (Matsubara et al., 2026).

These studies have not compared head size and helmet fit with brain noise included. The only helmet fitted at a constant gap (Zahran et al., 2022) matched the adult's closest scalp-to-sensor distance, not the adult's median gap. Whether the larger OPM advantage expected in children reflects head size itself or poor adult-helmet fit therefore remains untested. None of these studies checked its noise model against a measured covariance, and it is unclear how far a modeled comparison can be trusted while the OPM noise has not been measured.

Here, we compared the 306-channel Neuromag system with two OPM arrays in a realistic adult head and in smaller heads. The dense array was packed over the scalp; the site-matched array had one OPM at each Neuromag site. As the measure, we used the detectability of each cortical source by a matched filter that knows the source's field pattern and the covariance of the sensor, brain and room noise. We examined what determines the adult comparison and checked the noise model against Neuromag's measured noise. To separate head size from helmet fit, we placed smaller heads in the adult helmet and in a helmet fitted to each head at the adult's gap. We then asked whether the detectability difference carries over to the detection of simulated interictal spikes by idealized detectors. We also examined regional differences and spike localization as exploratory questions.

In this model, proximity alone did not decide the comparison. The array, OPM noise floor and SQUID helmet fit, which sets the distance of the SQUIDs from a smaller head, determined whether and by how much the on-scalp array had higher detectability. This result bears on comparisons of OPM and SQUID recordings in children and in epilepsy.

2. Materials and methods

2.1. Study design and estimands

To compare arrays without committing to a particular detector, we used the known-topography detectability of each source. This is the output SNR of a matched filter that knows the source's field pattern and the noise covariance,

$$d = \sqrt{\mathbf{s}^\top \mathbf{C}^{-1} \mathbf{s}}. \tag{1}$$

Here, $\mathbf{s}$ is the noise-free field of a 10-nAm cortical dipole at the array's channels. We treated the array's noise covariance, $\mathbf{C}$ (Section 2.4), as known, using the true covariance of the modeled noise. We compared arrays target by target using the ratio $d_{\mathrm{OPM}}/d_{\mathrm{Neuromag}}$ or its value in decibels,

$$D = 20 \log_{10} \frac{d_{\mathrm{OPM}}}{d_{\mathrm{Neuromag}}}, \tag{2}$$

which is positive when the OPM array has higher detectability.

We summarized each comparison over the cortex, taking for the adult the unweighted median of target-wise ratios over all 7,661 targets, including the 558 medial-wall vertices. Because the smaller heads differ in cortical area, we summarized each by the area-weighted median of D over its cortical targets, excluding the medial wall.

We set the adult's aggregation convention before adding the smaller heads and retained it rather than change it after the results were known. For the dense array, excluding the medial-wall vertices would change the adult's median ratio to 1.15; the smaller heads' area-weighted convention would give 1.16 (+1.27 dB).

To compare a smaller head with the adult, we placed both in the same helmet and pose (Section 2.3). We computed the change from the adult, Δ, from paired differences, rather than the difference of medians, with pairing determined by the anatomy. For the scaled adults, downscaled copies sharing the adult's vertices, we computed Δ as the median of $D_{\mathrm{head}}(v) - D_{\mathrm{adult}}(v)$ over common vertices, weighted by the smaller head's target areas. The other heads have their own cortical surfaces rather than the adult's vertices, so we first took the area-weighted median of D within each Desikan–Killiany atlas parcel containing at least 10 targets in both heads. In the 24- and 12-month templates, 66 and 65 parcels, respectively, met this criterion. We then computed Δ as the median of the parcel differences, weighted by the adult's parcel areas.

2.2. Head models

We used the MNE sample subject as the adult head (Gramfort et al., 2013), with its FreeSurfer surfaces and atlas labels (Desikan et al., 2006; Fischl, 2012). Targets were vertices of an MNE-Python oct-6 source space on the white-matter surface, each modeled as a 10-nAm dipole normal to the cortex. We retained only vertices at least 4 mm inside the inner skull and defined target depth as the distance to the nearest MRI scalp point. We computed forward fields with a three-layer boundary-element model (BEM; Hämäläinen and Sarvas, 1989), with conductivities of 0.3, 0.006, 0.3 S/m and the MRI scalp as its outer surface. We solved the BEM by linear collocation, using 5,120 triangles per skull surface and a scalp refined to 20,480 triangles.

We checked the BEM against an exact three-shell sphere and by refining the scalp. Relative to the exact sphere's field, the BEM field had a 95th-percentile error of 0.074 % at 2 mm and 0.038 % at 15 mm outside the outer surface. On a subset of targets, refining the scalp from 5,120 to 20,480 triangles changed the dense-array ratio from 1.149 to 1.141.

Four of the eight smaller heads form the basis of the main analysis (Table 1); we call them the principal smaller heads. Two were made by scaling the adult, one to school-age size and the other to the head circumference of the 24-month template. These scaled adults preserve the adult anatomy exactly and are therefore the main test of head size and helmet fit. The other two are the 24- and 12-month infant templates, averages of many MRIs (O'Reilly et al., 2021; Richards et al., 2016), which add infant head shapes to the analysis, subject to the scalp-convention caveat below. None of the anatomies is an independent replication, because all share the noise calibration, the adult's conductivities and the processing.

The remaining four heads are reported separately in Section S10. An MRI check classed the 18-month template as misregistered because its white surface fitted its T1-weighted image best after a 2.3-mm shift, compared with 0.5 mm in the other two templates. We obtained MRIs of three typically developing school-aged children, A, B and C (7.8–8.7 years), from OpenNeuro dataset ds005234 (Fadeev et al., 2024, 2025).

We prepared the children's anatomy in three further steps (Section S4). In this dataset, their inner skull lay a median of 0.8–2.5 mm below the scalp over the upper head. We therefore displaced it inward toward 8 mm below the scalp, keeping it at least 2 mm from the white-surface vertices. We then transferred the adult's fiducials to the children by a similarity transform fitted between the white surfaces and checked the children's surfaces against their T1-weighted MRIs (Fig. S6). The check found the surfaces registered to the images, but it cannot verify cortex within a few millimeters of the scalp. The 4-mm target limit also excluded 94–100 % of the children's cortex within 8 mm of the scalp, so their results do not describe a near-scalp cortex.

Further checks of the principal heads concerned scalp conventions and conductivity layers. The infant templates' scalps lie 1.3–1.6 mm outside their MRI head boundaries, whereas the adult's scalp lies 1.3 mm inside its boundary. The difference between the two conventions is 2.6–3.0 mm (Section S4.3). We also replaced the three-layer BEM with a single-compartment BEM, which has no skull or scalp conductivity layer. This changed the difference between each principal head's median D and the adult's by −0.02 to +0.01 dB. This check tests the conductivity layers, not the position of the scalp or inner skull.

2.3. Sensor arrays and helmets

We modeled the Neuromag system (306 channels; "Neuromag" below) using the sample recording's sensor geometry and MNE-Python's T3 coil definitions (types 3014 and 3024). We integrated each gradiometer at 8 points and each magnetometer at 16. We set Neuromag's white noise to the typical TRIUX datasheet values: 3.5 fT/√Hz for magnetometers and 3.6 fT/(cm √Hz) for gradiometers (Elekta Neuromag, n.d.). The coils were 18 mm outward from the helmet's inner surface (the Dewar spacing), and no signal-space separation was applied. We compared each OPM array with all 306 Neuromag channels and, as alternative comparators, with the 204 gradiometers or 102 magnetometers alone. Besides detectability, we computed peak-channel SNR and mean-power SNR (Goldenholz et al., 2009), the sensor-level measures defined in Table 2.

We modeled each OPM as a single-axis sensor with its sensing center nominally 7 mm above the MRI scalp (Alem et al., 2023; Osborne et al., 2018). We displaced sites outward if their sensor cells would otherwise have come within 1 mm of the head surface. In the adult, this affected 26 of the 208 dense sites and 11 of the 98 site-matched sites. We averaged each OPM's field over a 10-mm cube with a 27-point Gauss–Legendre rule, a modeling choice that changes the peak field

Table 1. Head models.

Head	Class	Head circumference (cm)	Cortex (cm^2)	Targets	OPM sites, dense / site-matched
Adult	MNE sample subject	58.6	1,878	7,103[a]	208 / 98
Principal smaller heads					
School-age size	scaled adult	52.5	1,470	6,949	174 / 89
2-year size	scaled adult	49.5	1,290	6,855	155 / 90
24-month template	infant template	49.5	1,062	7,559	151 / 83
12-month template	infant template	46.9	895	7,592	144 / 82
Reported separately (Section S10)					
18-month template[b]	infant template	49.1	975	7,363	157 / 80
Child A (7.8 years)	school-aged child[c]	52.0	1,852	7,138	155 / 90
Child B (8.3 years)	school-aged child[c]	48.6	1,666	6,866	153 / 85
Child C (8.7 years)	school-aged child[c]	53.5	1,869	7,392	167 / 91

Head circumference is that of the MRI scalp; cortex is the usable cortical area; targets exclude the medial wall. The median magnetometer-to-scalp gaps of each head in each helmet, with the fitted helmets' scale factors, are listed in Table S5. Infant templates: O'Reilly et al. (2021), from the Neurodevelopmental MRI Database (Richards et al., 2016). Children A–C: OpenNeuro ds005234 (Fadeev et al., 2024, 2025). [a]The adult's whole-cortex medians use all 7,661 targets, the medial wall included (Section 2.1). [b]Classed misregistered by the MRI check (Section 2.2). [c]Near-scalp anatomy not verifiable (Section 2.2).

by at most 1.9 %. The sensitive axis was the head-surface normal, averaged within 15 mm of the site. Had nearly radial axes been used instead, the dense-array ratio after projecting out the room field would have been 1.07 rather than 1.12 (Sections 2.4 and S3).

We set OPM white noise, flat over 1–40 Hz, to 15 fT/√Hz in the primary analysis and varied it over 7, 10, 15, 20 and 30 fT/√Hz. The primary level was a declared value rather than a device specification (Section S8.3).

Both OPM arrays were restricted to the same scalp region, above the plane through the preauricular points and a point 30 mm above the nasion. No OPM was within 20 mm of either preauricular point. Packing the dense array at a minimum spacing of 17 mm gave 208 sites on the adult, more than any clinical OPM array in Table S19. For the site-matched array, we placed one OPM at each Neuromag site whose scalp projection fell within this region: 98 of 102 sites (Fig. 1). Clinical systems reach about 100–128 channels with dual- or triaxial sensors at fewer sites (Feys et al., 2025; Shen et al., 2026). We did not model these arrays. Neither single-axis array represents them, because both differ from clinical arrays in site count and sensing axes. To test the effect of extra axes, however, we added a triaxial array at the site-matched sites (294 channels).

In the fixed adult helmet, we used the adult's sensors and noise unchanged for every head (Fig. 2). For the primary placement, we positioned each head's coordinate frame where the adult's had been recorded, then raised the head until the nearest magnetometer coil was 20 mm from the scalp. We call this placement top contact, a best-fit assumption that favors Neuromag (Gaetz et al., 2008; Tesan et al., 2010). To describe the effect of placement, we also used 12 source-blind placements chosen from the geometry alone (contacts, shifts and rotations; Section S5). In contrast, we rebuilt the OPM arrays on every smaller head, using the same coverage and spacing rules for the dense array. The site-matched array used that head's Neuromag sites at top contact (Table 1).

To separate head size from helmet fit, we also placed each head in a helmet fitted at the adult's gap. We translated the head laterally from the recorded pose to the helmet midline without raising it to contact (the laterally centered pose). We then scaled the helmet about the head origin until the median magnetometer-to-scalp gap matched the adult's in the same pose, 29.6 mm. For comparison, the adult's gap is 29.8 mm at its measured head position and 28.4 mm at top contact. No coil could come within the 18-mm Dewar spacing of the scalp; no principal head reached this limit. The fitted helmet retains the adult's sensors and therefore does not model a pediatric SQUID system.

Our primary measure of helmet fit is the paired change, from the adult to a smaller head, in how much detectability Neuromag loses in the fixed helmet relative to the helmet fitted at the adult's gap. We call this measure the interaction. For each target, the contrast between helmets within one head is

$$F(v) = D_{\text{fixed}}(v) - D_{\text{fitted}}(v) = 20 \log_{10} \frac{d_{\text{Neuromag, fitted}}(v)}{d_{\text{Neuromag, fixed}}(v)}, \tag{3}$$

in which OPM detectability cancels. Because the fitted helmet is built about the laterally centered pose rather than at contact, the contrast F combines changes in helmet scale and pose. We obtained the interaction for a smaller head by applying the Δ estimator of Section 2.1 to F instead of D. For the scaled adults, this is the median of $F_{\text{head}}(v) - F_{\text{adult}}(v)$ over common vertices, weighted by the smaller head's target areas. For the templates, it is the median of parcel differences, weighted by the adult's parcel areas.

As a median of paired differences, the interaction equals neither the difference of the two median contrasts nor $\Delta_{\text{fixed}} - \Delta_{\text{fitted}}$. Table S6 gives each head's non-additivity residual. The interaction does not depend on the assumed OPM noise, because the detectability of the OPM array, which is the same in both helmets, cancels in F. It does, however, depend on the modeled Neuromag noise and on the anatomy.

We used three secondary controls of the construction (Section S5). We fitted one helmet at the adult's top-contact gap, 28.4 mm, and scaled another with the head, thereby scaling the gap as well. To compare heads at equal OPM site count, we kept the helmet fitted at the adult's gap unchanged and subsampled the adult's dense array to each smaller head's site count.

2.4. Noise model, validation and sensitivity analyses

We exposed all arrays to the same noise sources in a band of 1–40 Hz. The noise comprised white sensor noise, a cortical background of spontaneous brain activity, and the field of distant room sources (the room field). We computed each array's noise covariance in closed form and treated it as known in the detectability measure (the true covariance),

$$\mathbf{C} = \mathbf{\Sigma}_{\text{n}} + \sigma^2 \mathbf{L}\mathbf{A}\mathbf{L}^{\top} + \mathbf{E}\mathbf{\Sigma}_{\text{e}}\mathbf{E}^{\top}. \tag{4}$$

Here, $\mathbf{\Sigma}_{\text{n}}$ is the diagonal covariance of white sensor noise. In the brain-noise term, $\mathbf{L}$ is the lead-field matrix of 1,755 independent cortex-normal background dipoles on a 7-mm grid, and the diagonal matrix $\mathbf{A}$ contains their Voronoi-cell areas. In the room-field term, $\mathbf{E}$ contains the array's responses to eight external-field terms (three uniform and five linear-gradient components), and $\mathbf{\Sigma}_{\text{e}}$ is the covariance of their coefficients. The background is a cortically constrained form of the random-dipole model of spontaneous activity (de Munck et al., 1992). Each background dipole has a moment variance of σ^2 times its area ($\sigma^2 = 0.0354$ nAm^2 per mm^2). As for the targets, we omitted cortex within 4 mm of the inner skull from the background.

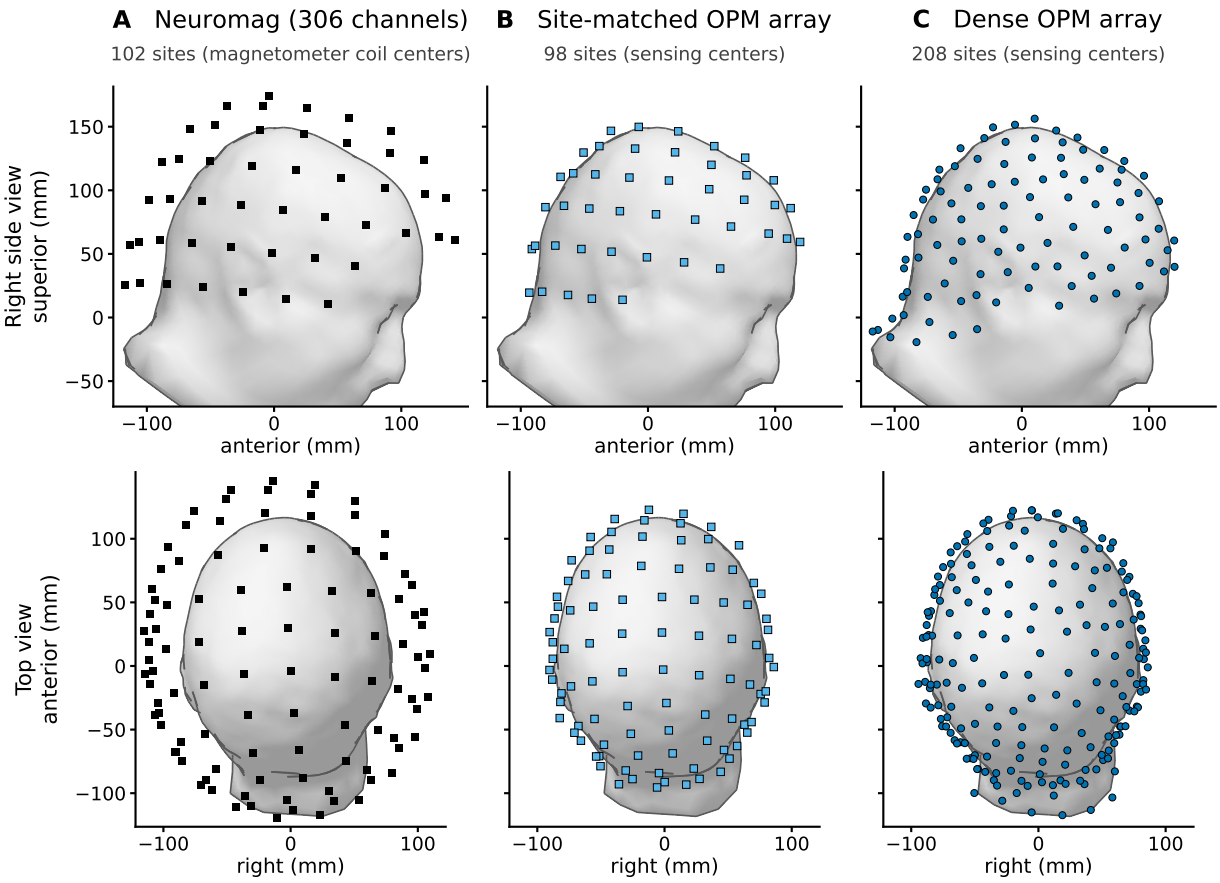


Fig. 1. Sensor arrays on the adult head (MNE sample subject), viewed from the right (sensors on the right half of the head) and from above; sensors hidden behind the head are not drawn. Head: the outer boundary-element surface of the forward model (conformed to the MRI scalp). (A) Neuromag (306 channels; black squares): 102 sites, each holding one magnetometer and two planar gradiometers, at the adult's measured head position (median 29.8 mm from the scalp). (B) Site-matched OPM array (light-blue squares): the 98 Neuromag sites inside the OPM coverage region, projected onto the scalp. (C) Dense OPM array (blue circles): 208 sites at least 17 mm apart. OPM sensing centers lie nominally 7 mm from the scalp and measure the field normal to the head surface. The two OPM arrays keep these colors in Figs. 3–5 and 7.

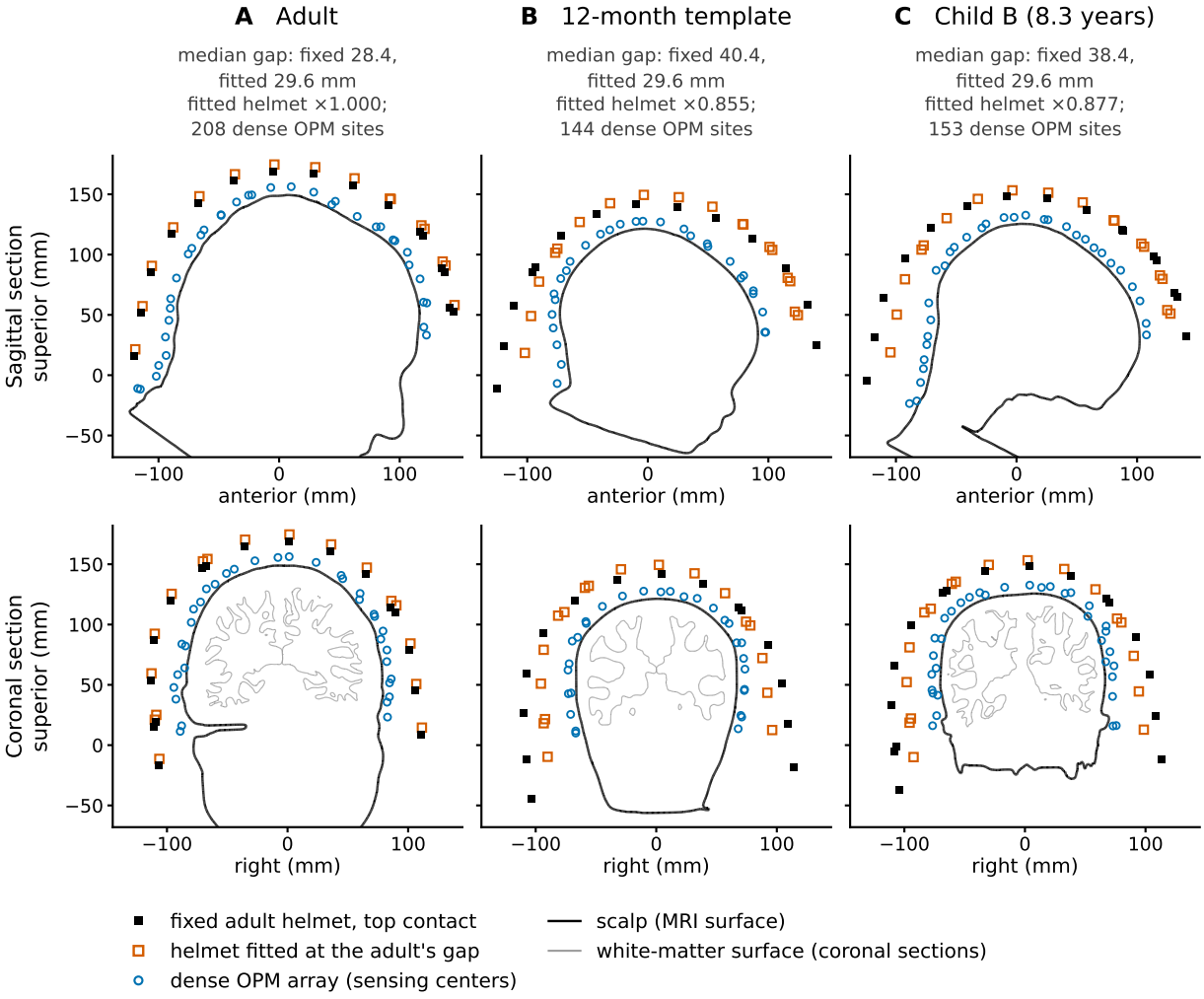


Fig. 2. The fixed and the fitted helmet. Sagittal and coronal sections through the head origin of (A) the adult, (B) the 12-month template and (C) child B (one of the heads of Section S10), at one scale: scalp (black; the outer contour of each section, cropped where it extends below the panel; the 12-month template's contours close along the lower boundary of its MRI, and child B's coronal contour closes within the panel) and, in the coronal sections, the white-matter surface on which the sources lie (gray); filled black squares, the Neuromag magnetometers of the fixed adult helmet at top contact; open vermilion squares, the helmet fitted at the adult's gap, built about the laterally centered pose (Section 2.3; in the adult it differs from the fixed helmet in pose only), with its scale factor above each column; open blue circles, the dense OPM sites; only sensors within 20 mm of the section are drawn. Median magnetometer-to-scalp gap: 28.4, 40.4 and 38.4 mm in the fixed helmet; 29.6 mm in the fitted helmet. The children's modeled skull is shown against their MRIs in Fig. S6A. Infant template: O'Reilly et al. (2021), from the Neurodevelopmental MRI Database (Richards et al., 2016); child B: OpenNeuro ds005234 (Fadeev et al., 2024, 2025).

We fitted two quantities to the sample dataset: the room-field covariance $\mathbf{\Sigma}_e$, to its empty-room recording, and σ^2, set so that the modeled brain noise of the median good gradiometer matched the measured level, 37.1 fT/cm root-mean-square. Everything else, apart from both systems' datasheet sensor noise, is predicted rather than fitted. These predictions come from the same recording, so the checks below test the model within that recording, not in another subject or session.

We analyzed four noise conditions: sensor noise only; sensor plus brain noise, the primary condition; sensor, brain and room noise; and the room field projected out. To represent interference suppression, we derived the last condition from the sensor, brain and room-noise condition by removing the room field from every array with one projection. This removes the array's responses to the room-field terms after weighting each channel by its inverse sensor-noise variance (Eq. (S1)). We designated this condition, the room field projected out, as the secondary condition before any simulation.

To check the model, we compared its Neuromag covariance with the measured covariance and recomputed Neuromag's detectability using the measured one (Section S3). We band-passed the recording to 1–40 Hz and excluded one bad channel. The measured covariance was the empirical covariance of 320 pre-stimulus windows (–0.2 to 0 s; 38,400 samples), without regularization or projections. We took its difference from the empty-room covariance (63,666 samples) as the measured brain noise. To remove the inflation in detectability from estimating covariance with finite data, we generated 40 phase-randomized surrogates of the whole recording, preserving its spectra and cross-spectra. We estimated their covariance from the same samples and divided each target's measured detectability by the resulting inflation (median factor 1.033).

Because the OPM covariance cannot be measured here, we computed the implied OPM/Neuromag ratio under five assumptions about how each system's real noise departs from the model. In scenario A, both systems share the model's error, leaving the modeled ratio unchanged. Scenario B scales the variances of both cortical backgrounds by $k = 1.86$, the factor that matches the modeled magnetometer brain-noise variance to the measured one. The remaining scenarios replace all or part of Neuromag's modeled noise with measured noise. Scenarios C and D use Neuromag's measured covariance: the variance of the OPM background is scaled by k in C, whereas the OPM retains its modeled covariance in D. Scenario E combines Neuromag's measured empty-room noise with its modeled brain noise.

In each sensitivity analysis, we changed one element of the adult model and recalibrated σ^2. We added cortex 2–4 mm from the inner skull, termed the near-skull cortex, to the background, the targets, or both. We also tested colored OPM noise ($1/f$ plus white noise, with corners at 1, 3 and 10 Hz) under a matched filter that also whitens across frequency. Separate analyses added idealized cardiac and ocular sources. The joint runs added the near-skull cortex to the background and gave the OPM noise a 10-Hz corner under the frequency-whitening detector, with and without the cardiac and ocular sources. We did not model noise specific to OPMs, such as cross-talk between neighboring sensors and errors in gain and sensitive axis. The colored-noise variants alone represented the distance-independent low-frequency noise of a real OPM (Section S3.2).

2.5. Simulated interictal spikes

We designed the spike simulations to ask whether the detectability difference carries over to transient-event detection by an idealized detector under the modeled noise. These simulations do not test the noise model. We ran the noise model in the time domain, using segments shared by all arrays, and stretched the spike-wave complex of Hunold et al. (2016) to three morphologies. We analyzed the data with two matched-filter detectors, both whitened using a Ledoit–Wolf covariance (Ledoit and Wolf, 2004) estimated from 10 min of null data, rather than with the true covariance.

The oracle detector knew the source and time and tested at a per-trial false-positive probability of 0.001. The practical detector knew neither and scanned candidate field patterns on a 10-mm grid that did not contain the true sources. We fixed its thresholds in advance (frozen) on independent null data for a nominal 1 false event per minute. Both detectors are idealized because they use the true forward model and the simulated waveform family. We interpolated S50, the dipole strength detected in 50 % of events, linearly in log strength between tested strengths. In both spike runs, the smaller heads were at top contact in the fixed adult helmet, whereas the adult was at its measured head position. At this position, the adult's area-weighted D is 0.28 dB higher than at top contact.

In the exploratory run, we placed spikes at 72 locations per anatomy, stratified by orientation and depth, with 18 locations in each of four depth bands, ranging from 10–20 to 45–70 mm. At each location, we simulated 18 focal events of 10–320 nAm and patch events from cortical patches of 10-mm radius. Each event had its own noise realization and was identical for every array. All comparisons were within an anatomy; the anatomies are not ranked against one another. We also localized spikes at 24 locations by equivalent-current-dipole fits and dynamic statistical parametric mapping (dSPM; Dale et al., 2000) at the true peak sample. Localization used a single-compartment BEM and a coregistration error of 2 mm and 2°. Paired comparisons of localization error included every event, detected or not; a joint measure also required detection (Section S7).

We then pre-specified a second run. Its endpoint compared the dense OPM array with Neuromag for focal spikes 10–20 mm below the scalp in every anatomy, under the practical detector at a nominal 1 false event per minute. We drew 36 new locations per anatomy and five noise replicates per event, with the first replicate carrying the endpoint. Separate null data were used for threshold fitting and rate evaluation. The threshold was set on 20 min of calibration null data and rests on about 20 false events, corresponding to a Poisson uncertainty of about 22 %. The oracle detector's threshold does not depend on this calibration.

The declared statistic was a two-sided sign-flip test on per-location differences in detection counts, Holm-adjusted over the anatomies (Holm, 1979) at $\alpha = 0.05$. The paired S50 ratio r served as the effect size; a value of r means that the OPM array detects a spike at a strength $100\,(1-1/r)$ % lower than Neuromag. The declared test estimated p by Monte Carlo when more than 20 location differences were nonzero. We instead report exact p values over all sign patterns, which give the same decisions in all 25 families of Holm-adjusted tests (Section S7). As secondary analyses, we declared a mismatched detector using off-template morphologies, candidate patterns from a single-compartment BEM and a shared coregistration error of 2 mm and 2°. Further secondary analyses covered the site-matched array, the oracle detector and thresholds re-fitted on held-out null data.

2.6. Statistical analysis

To preserve dependence between neighboring targets within a parcel, we computed intervals over targets as 95 % percentile bootstrap intervals that resample atlas parcels as units: the adult's 70 labels or each smaller head's parcels. We used 1,000 resamples for primary comparisons, noise-model sensitivity analyses and the measured-covariance check, and 200 for the other adult sensitivity analyses and some secondary constructions of the smaller heads (Table S16). For Δ and the interaction, the medians of paired differences between a smaller head and the adult, we resampled parcels together with their paired differences. In the spike study, locations were the units of resampling and testing.

Within one anatomy and noise model, the parcel intervals describe the sensitivity of a whole-cortex median to the parcels it pools. They are neither population confidence intervals nor measures of the spread of effects across the cortex, and they exclude the larger uncertainties of model

Table 2. Main estimands.

Estimand	Definition	Unit; interval	Section
Adult ratio	Median over all 7,661 targets (medial wall included) of the target-wise ratio $d_{\mathrm{OPM}}/d_{\mathrm{Neuromag}}$	ratio; parcel bootstrap	3.1–3.3
Scenario ratio	The adult ratio with sensor, brain and room noise (Neuromag's 305 channels, one bad channel excluded) under one of five assumptions about how each system's real noise departs from the model (scenarios A–E)	ratio; parcel bootstrap	3.2
Break-even noise	OPM white noise at which the adult ratio equals 1, under the modeled Neuromag noise	fT/√Hz; parcel bootstrap	3.1, 3.2
Δ	Area-weighted median of the paired differences in D between a smaller head and the adult, each in the same helmet and pose: per vertex for the scaled adults, per parcel (weighted by the adult's parcel areas) otherwise	dB; parcel bootstrap	3.4, 3.5
Interaction[a]	The Δ estimator applied to the within-head contrast between the fixed and the fitted helmet, F of Eq. (3); a paired median, not $\Delta_{\mathrm{fixed}} - \Delta_{\mathrm{fitted}}$	dB; parcel bootstrap	3.5
Peak-channel SNR ratio	Each system's largest ratio of a channel's noise-free signal to its noise standard deviation, OPM over Neuromag	ratio; parcel bootstrap	3.1, 4.3
Mean-power SNR ratio	Channel mean of the squared SNRs (Goldenholz et al., 2009), OPM over Neuromag, as an amplitude ratio	ratio; parcel bootstrap	3.1
S50 ratio	Dipole strength for 50 % detection, Neuromag over OPM, from the location-pooled detection curves	ratio; location bootstrap	3.6

d: known-topography detectability with the true covariance, Eq. (1); D: Eq. (2). Neuromag: all 306 channels unless stated. Top contact: the primary placement of the fixed adult helmet; fitted helmet: the helmet fitted at the adult's gap (Section 2.3). At the adult's measured head position, the dense array's ratio is 1.14 (D = +1.17 dB); Section 2.1 gives it under the conventions of the smaller heads. [a]Independent of the assumed OPM noise; it depends on the modeled Neuromag noise and on the anatomy.

error, head placement and between-subject variation. Whether an interval includes 1 (or 0 dB) is therefore reported descriptively, not as a test. Table 2 lists the main estimands, and Table S2 lists all of them.

Several choices were made after results were known. After the whole-cortex results were known, we named the primary helmet placement, the regions of interest and the exploratory spike endpoint, chosen from 24 comparisons per OPM array. We later added the helmet fitted at the adult's gap, the measured-covariance check with its five scenarios and the sensitivity analyses; these additions did not change the model. The model itself, however, was revised while results were known (Section S8.2).

We declared the pre-specified run in a configuration file committed to the public code repository the day before the run, which used the same code and configuration. Only the repository history attests this; the run was not registered externally. We chose the number of locations, double the exploratory 18 per band, after two pilot runs in child A. Their outputs were not retained, and their seeds were not reused.

2.7. Software

We ran the simulations with MNE-Python 1.13.2 (Gramfort et al., 2013) and NumPy 2.5.3 (Harris et al., 2020); Table S17 lists the random seeds. The analysis code was written and run with the assistance of a generative AI tool (Claude, Anthropic), as declared at the end of the article. Automated tests in the public repository cover the code's numerical components.

3. Results

3.1. Determinants of the adult comparison

Brain noise reversed the comparison between the dense OPM array and Neuromag (Fig. 3; Table 3). With sensor noise alone, Neuromag had higher detectability than either OPM array (Fig. 3). When brain noise was added, the dense array had the higher detectability at 7,657 of the 7,661 cortical targets, with a median ratio to Neuromag of 1.14 [1.12, 1.17] over all targets. The site-matched array, however, did not differ clearly from Neuromag (1.01 [0.99, 1.03]). Projecting out the room field from every array gave the same pattern, although the site-matched array's ratio was then slightly below parity.

Brain noise dominated both systems' noise (Section S3). With the room field included, brain noise accounted for 92 % of an OPM channel's in-band variance in the dense array and 74 % of a Neuromag magnetometer's. The room field accounted for most of the remaining magnetometer variance (25 %). Brain noise thus raised Neuromag magnetometer noise about 9.3-fold above its sensor floor, compared with only about 5.7-fold for an OPM. The higher OPM sensor noise therefore mattered less once brain noise was present.

The Neuromag channels used as comparator and the SNR measure also affected the ratio (Fig. 3; Table 3). The dense array's advantage was larger against the gradiometers alone than against the magnetometers alone (Table 3). Peak-channel SNR, the closest analog of sensor-level SNR in clinical studies (Section S1), favored the dense array against the magnetometers, whereas it favored Neuromag when Neuromag's best channel of all 306 was used. Both OPM arrays had only small advantages in mean-power SNR (Fig. 3). Thus, which system a reported SNR favors depends on the Neuromag channels and SNR measure used.

Among the model variants, the OPM white-noise floor had the largest effect on the ratio (Fig. 4D; Table 3). As it increased over the tested range, the dense-array ratio decreased to parity, whereas the site-matched ratio fell below it (Fig. 4D). Under the modeled noise, the break-even OPM noise of the dense array, the OPM white noise at which its ratio equals 1, was 31.5 [28.5, 34.9] fT/√Hz (Fig. 4D, diamonds). The site-matched array broke even at 16.7 [14.0, 19.4] fT/√Hz, an interval that includes the primary OPM noise level. With the room field projected out, the dense array's break-even level fell to 26.0 [22.3, 29.5] fT/√Hz (Fig. 4D, open diamonds), and the site-matched array's fell to 8.9 [5.6, 12.4] fT/√Hz. All these levels assume the modeled Neuromag noise, which the next section compares with measured noise.

The other model variants changed the dense-array ratio less over the ranges tested (Fig. 3; Table 3). An additional OPM-to-scalp gap reduced the ratio without reversing it. Combining an OPM noise of 30 fT/√Hz with an additional 3-mm gap did reverse it (Fig. 3), giving a ratio of 0.97 [0.96, 0.98]. The triaxial array at the site-matched sites recovered most of the dense array's advantage, and still recovered part of it when its tangential axes were twice as noisy (Fig. 3, triangles). The ratio changed little with a covariance estimated from as many simulated samples as a minute of data would contain, or with a 3–70-Hz band instead of the primary band. Thus, besides brain noise, array design and the OPM noise floor determined which system had the higher detectability in the adult; the comparator and SNR measure determined which system a reported SNR favored.

3.2. The noise model against measured Neuromag noise

The noise model reproduced some properties of Neuromag's measured noise and missed others (Figs. S2 and S3). It predicted 0.73 of the measured magnetometer brain-noise amplitude, which was not used in calibration, and 0.89 with the heart's field removed. Calibrating the brain-noise scale on a random half of the sites gave agreement with the other half's measured gradiometer level nearly by construction (1.00; Fig. S3B). Across 1,000 random splits, the 95 % range of

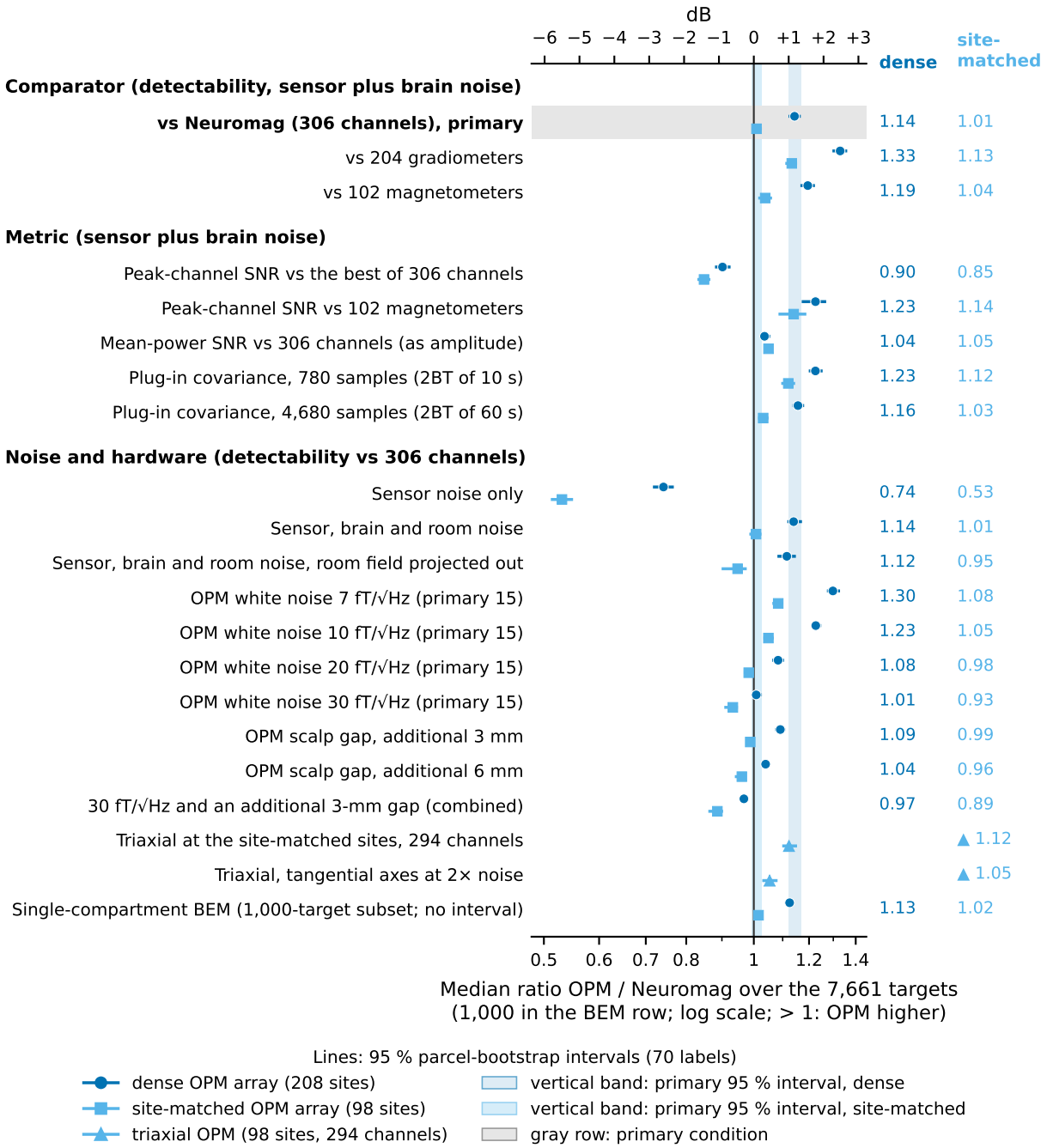


Fig. 3. Determinants of the adult comparison. Median over the 7,661 targets (1,000 in the single-compartment BEM row) of the target-wise ratio OPM/Neuromag, in the metric and against the comparator of each row (detectability against all 306 channels unless stated), with 95 % parcel-bootstrap intervals over 70 atlas labels (1,000 resamples; 200 for the OPM-noise, gap and combined rows; none for the BEM row); logarithmic axis, with the same ratio in dB on the top axis (D for the detectability rows). The gray row and the vertical bands mark the primary condition, over all targets, and its intervals: detectability with the true covariance, sensor plus brain noise, OPM noise of 15 fT/√Hz, no gap beyond the nominal OPM standoff, three-layer BEM and the adult at its measured head position. Each other row changes one element of it, except the peak-channel SNR against the magnetometers (metric and comparator), the combined row (30 fT/√Hz with an additional 3-mm gap) and the BEM row (head model and targets; on its subset, the three-layer BEM gives 1.14 and 1.01). Plug-in covariances: 780 or 4,680 independent samples (2BT, twice the bandwidth B times the duration T, for T = 10 and 60 s). Right-hand columns: the medians; Table 3 lists all rows but the 780-sample and BEM rows, with their intervals.

Table 3. The adult comparison under each noise condition, comparator, SNR measure, model variant, noise scenario and noise-model omission.

Condition	Dense OPM array	Site-matched OPM array
Noise condition		
Sensor noise only	0.74 [0.72, 0.77]	0.53 [0.51, 0.55]
Sensor plus brain noise (primary)	1.14 [1.12, 1.17]	1.01 [0.99, 1.03]
Sensor, brain and room noise	1.14 [1.12, 1.17]	1.01 [0.99, 1.03]
Sensor, brain and room noise, room field projected out	1.12 [1.08, 1.15]	0.95 [0.90, 0.98]
Comparator and SNR measure		
Against the 204 gradiometers	1.33 [1.30, 1.36]	1.13 [1.11, 1.15]
Against the 102 magnetometers	1.19 [1.17, 1.22]	1.04 [1.01, 1.06]
Peak-channel SNR, against the magnetometers	1.23 [1.17, 1.27]	1.14 [1.08, 1.19]
Peak-channel SNR, against all 306 channels	0.90 [0.88, 0.93]	0.85 [0.83, 0.87]
Mean-power SNR	1.04 [1.02, 1.06]	1.05 [1.04, 1.06]
Model variants		
OPM noise 7 fT/√Hz[c]	1.30 [1.27, 1.33]	1.08 [1.06, 1.10]
OPM noise 10 fT/√Hz[c]	1.23 [1.21, 1.25]	1.05 [1.03, 1.07]
OPM noise 20 fT/√Hz[c]	1.08 [1.06, 1.11]	0.98 [0.97, 1.00]
OPM noise 30 fT/√Hz[c]	1.01 [0.99, 1.03]	0.93 [0.91, 0.95]
Break-even OPM noise (fT/√Hz)	31.5 [28.5, 34.9]	16.7 [14.0, 19.4]
Additional OPM gap of 3 mm[c]	1.09 [1.07, 1.11]	0.99 [0.97, 1.00]
Additional OPM gap of 6 mm[c]	1.04 [1.03, 1.05]	0.96 [0.94, 0.97]
30 fT/√Hz and an additional 3-mm gap[c]	0.97 [0.96, 0.98]	0.89 [0.86, 0.90]
Plug-in covariance, 4,680 samples (60 s)	1.16 [1.14, 1.18]	1.03 [1.01, 1.05]
3–70 Hz band[a]	1.14	1.01
Triaxial OPMs at the site-matched sites	–	1.12 [1.10, 1.15]
tangential axes twice as noisy	–	1.05 [1.03, 1.08]
Noise scenarios (sensor, brain and room noise)		
A: both systems share the error of the model	1.14 [1.12, 1.17]	1.01 [0.98, 1.03]
B: both background variances scaled by k	1.14 [1.12, 1.17]	1.00 [0.98, 1.02]
C: Neuromag measured, OPM background variance scaled by k	0.80 [0.76, 0.84]	0.68 [0.65, 0.70]
D: Neuromag measured, OPM modeled	1.03 [0.99, 1.08]	0.89 [0.86, 0.92]
E: measured empty-room noise of Neuromag, modeled brain noise	1.13 [1.10, 1.16]	1.00 [0.98, 1.02]
Noise-model omissions		
Cortex 2–4 mm from the inner skull as background	1.14 [1.12, 1.16]	1.01 [0.99, 1.02]
as additional targets[b]	1.15 [1.13, 1.18]	1.01 [0.99, 1.03]
10-Hz 1/f corner, primary detector	1.07 [1.05, 1.10]	0.98 [0.96, 0.99]
frequency-whitening detector	1.08 [1.06, 1.11]	0.98 [0.96, 1.00]
Joint run: near-skull background, 10-Hz corner, frequency-whitening detector	1.08 [1.06, 1.11]	0.98 [0.96, 1.00]
with cardiac and ocular sources	1.09 [1.06, 1.12]	0.97 [0.94, 0.99]

Median over the 7,661 targets of the target-wise ratio OPM/Neuromag, of detectability unless another metric is stated (all 306 channels unless stated), with the adult at its measured head position and an OPM white noise of 15 fT/√Hz unless stated; brackets, 95 % parcel-bootstrap intervals. The noise is sensor plus brain noise unless stated, and the scenarios include the room field. Break-even OPM noise: the OPM white noise at which the ratio equals 1 under the modeled Neuromag noise. The scenarios use the 305 good channels of Neuromag, and k is the factor by which the variance of a cortical background is scaled so that the modeled magnetometer brain-noise variance matches the measured one (Section 2.4). Plug-in covariance: estimated from as many simulated independent samples as 60 s of data hold in the band. Cardiac and ocular sources alone changed the ratio of the dense array by −1.6 % to +1.6 %. [a]Interval not shown: it was computed by resampling targets, not parcels. [b]Median over the 7,661 targets and the 463 near-skull targets. [c]Interval from 200 resamples; the other intervals from 1,000 (Table S16).

this model-to-measured ratio was [0.93, 1.09]. However, the modeled brain noise was too flat from inferior to superior sites. The model calibrated on the inferior half predicted 1.34 times the measured gradiometer amplitude in the superior half; calibrated on the superior half, it predicted 0.75 times that in the inferior half. The model also predicted per-channel variances poorly, and its magnetometer correlations decreased with distance faster than measured correlations. Thus, the measured covariance differed from the modeled one in structure as well as in level.

The measured brain noise also contained variance that the model lacks. This excess, the positive part of the difference between the measured and modeled magnetometer brain-noise covariances, summed to 1.10 times the summed modeled brain-noise variance. Sampling error alone would produce an excess of 0.02–0.05 times that variance. Nevertheless, Neuromag's detectability was higher with its measured covariance than with its modeled covariance (Fig. 4A). The median ratio of these two Neuromag detectabilities (measured over modeled covariance) was 1.14 [1.08, 1.20], as large as the dense array's entire advantage. It varied by lobe, from 0.89 in the temporal lobe to 1.40 in the cingulate. This higher detectability may appear paradoxical, but detectability depends on the structure of the covariance, not on its total variance alone. Despite greater channel-level noise, the measured covariance gave higher detectability at 73 % of targets, so it cannot exceed the modeled covariance in every spatial direction. Its excess lay in a few far-field components that whitening down-weights (Section S3).

What the measured Neuromag noise implies for the OPM arrays depends on the real OPM noise, which no measurement here constrains (Fig. 4B; Table 3). Both arrays' ratios were nearly unchanged under shared model error (scenario A) or with the variances of both cortical backgrounds scaled to match the measured magnetometer brain noise (scenario B). Using Neuromag's measured empty-room noise with modeled brain noise also left the ratios nearly unchanged (scenario E). Scenario C was least favorable to OPMs: Neuromag's measured covariance and the OPM background scaled as in scenario B lowered both arrays' ratios below parity. With Neuromag's measured covariance and OPM noise unchanged from the model (scenario D), the dense-array ratio decreased to 1.03 [0.99, 1.08], an interval that includes 1 (Fig. 4B). The site-matched ratio fell below parity, to 0.89 [0.86, 0.92]. Thus, under scenario D, neither array had a clear advantage over Neuromag at the primary OPM noise level.

We did not compute break-even levels under scenario D, but their direction follows from these ratios. Neuromag's measured detectability is independent of OPM noise, so the scenario-D ratio, like the modeled ratio, falls as OPM noise rises. At the point estimate of 1.03 at 15 fT/√Hz, the dense array would break even above that noise level, although its ratio interval includes 1. The site-matched array, with a ratio of 0.89, would break even below that level, if at all within the tested range. The medians alone cannot determine how far either level would move from its modeled value, because the measured-noise correction varies across targets.

The tested model omissions changed the dense array's advantage little at the primary OPM noise (Fig. 4C; Table 3). Adding the near-skull cortex as background or as additional targets changed the ratio little (Fig. 4C). Colored OPM noise at the highest tested corner frequency, 10 Hz, slightly reduced the ratio under both the primary and frequency-whitening detectors. Adding cardiac and ocular sources, separately or together, likewise changed the ratio by only −1.6 % to +1.6 %.

The joint runs combined the near-skull background and a 10-Hz OPM-noise corner under the frequency-whitening detector (Fig. 4D, dash-dotted). With white OPM noise at 15 fT/√Hz, this detector gave a dense-array ratio of 1.13 [1.10, 1.16] (Fig. 4C). The joint run reduced it to 1.08 [1.06, 1.11] (Table 3). Raising white OPM noise alone to 20 fT/√Hz reduced it more, to 1.06 [1.03, 1.08]. With both changes, the ratio was close to parity at 20 fT/√Hz (1.016 [0.997, 1.035], or 1.02 [1.00, 1.05] with the cardiac and ocular sources added) and below it at 30 fT/√Hz

(0.93; Fig. 4D). Under the same detector, the omitted noise sources thus lowered the ratio less than raising the OPM noise from 15 to 20 fT/√Hz did.

3.3. Depth and brain regions

The advantage of the dense array was largest near the scalp and decreased with depth (Fig. 5A). Its ratio decreased from 1.60 [1.55, 1.65] in the 10–15 mm depth bin to 1.13 [1.12, 1.14] in the 25–30 mm bin. It then changed little, remaining at 1.05–1.07 from the 35–40 to the 55–60 mm bin. Even the deepest targets thus retained a small advantage, and every interval in these bins excluded parity. The site-matched array, in contrast, had an equal-SNR depth, as in the spherical analysis, even with sensor plus brain noise. Its intervals were above 1 down to the 25–30 mm bin and below 1 from the 40–45 mm bin, so that Neuromag had the higher detectability for the deeper sources.

Projecting out the room field reduced the ratios of both OPM arrays at depth (Fig. 5A, dashed). The dense array then also had an equal-SNR depth, with bin medians below 1 from the 45–50 mm bin, although its intervals were entirely below 1 only in the 55–60 mm bin. The equal-SNR depth of the site-matched array became shallower, with bin medians below 1 from the 25–30 mm bin and intervals below 1 from the 30–35 mm bin. Whether an OPM array keeps its advantage at depth thus depends on how the room field is treated.

The ratio of the dense array decreased with depth whether or not brain noise was present. The decrease was already present with sensor noise alone (Fig. 5A, dotted) and in the noise-free peak field (Fig. 5B). Adding brain noise multiplied the ratio by a nearly constant factor in every bin (1.51–1.68). Together, these observations suggest that the decrease arose from the source–sensor geometry rather than from the noise.

The dense array had the higher detectability almost everywhere on the cortex, most of all on the lateral convexity (Fig. 6A). The regional results, however, are exploratory point estimates under the modeled noise. Among the lobes, the advantage of the dense array was largest in the frontal lobe (1.22 [1.18, 1.26]; Fig. 7A). The smallest advantages were found in the cingulate (1.05 [1.04, 1.07]) and the insula (1.09; Fig. 7A). Of the regions chosen from the epilepsy literature, the precentral gyrus favored the dense array most (1.21) and the parahippocampal cortex least (1.12; Fig. 7A). However, another 12 atlas parcels, mostly cingulate and medial, had a smaller advantage than the parahippocampal cortex.

The site-matched array had its lowest ratios on ventral and polar surfaces (Fig. 6B). Its ratio was 0.71 at the frontal pole and 0.83 at the temporal pole. In the medial and lateral orbitofrontal cortex, it was 0.78 and 0.79, respectively. Among the regions from the literature, the ratio of the site-matched array was lowest in the mesial temporal cortex, the parahippocampal and entorhinal parcels (0.92).

With Neuromag's measured noise (scenario D), however, the dense array's regional ranking changed. Its ratio became 1.32 in the temporal lobe, the highest of any lobe, and 1.01 in the frontal lobe. The ratio fell below 1 in the cingulate (0.75) and the parietal lobe (0.96). Under scenario D, the depth-bin medians of the dense array were also below 1 from the 25–30 mm bin. Overall, the regional ranking depended on the noise model as well as on the array.

3.4. Smaller heads in the fixed adult helmet

Every principal smaller head showed a larger OPM advantage than the adult in the fixed adult helmet (Fig. 8; Table 4). At top contact, their D values were +1.50 to +2.13 dB, compared with +1.00 [+0.82, +1.17] dB in the adult (Fig. 8A). At this placement, the paired change from the adult, Δ, was +0.44 to +1.07 dB (Fig. 8B). Halving or doubling cortical background variance per unit area left every principal head's D above the adult's. The excess was +0.53 to +1.18 and

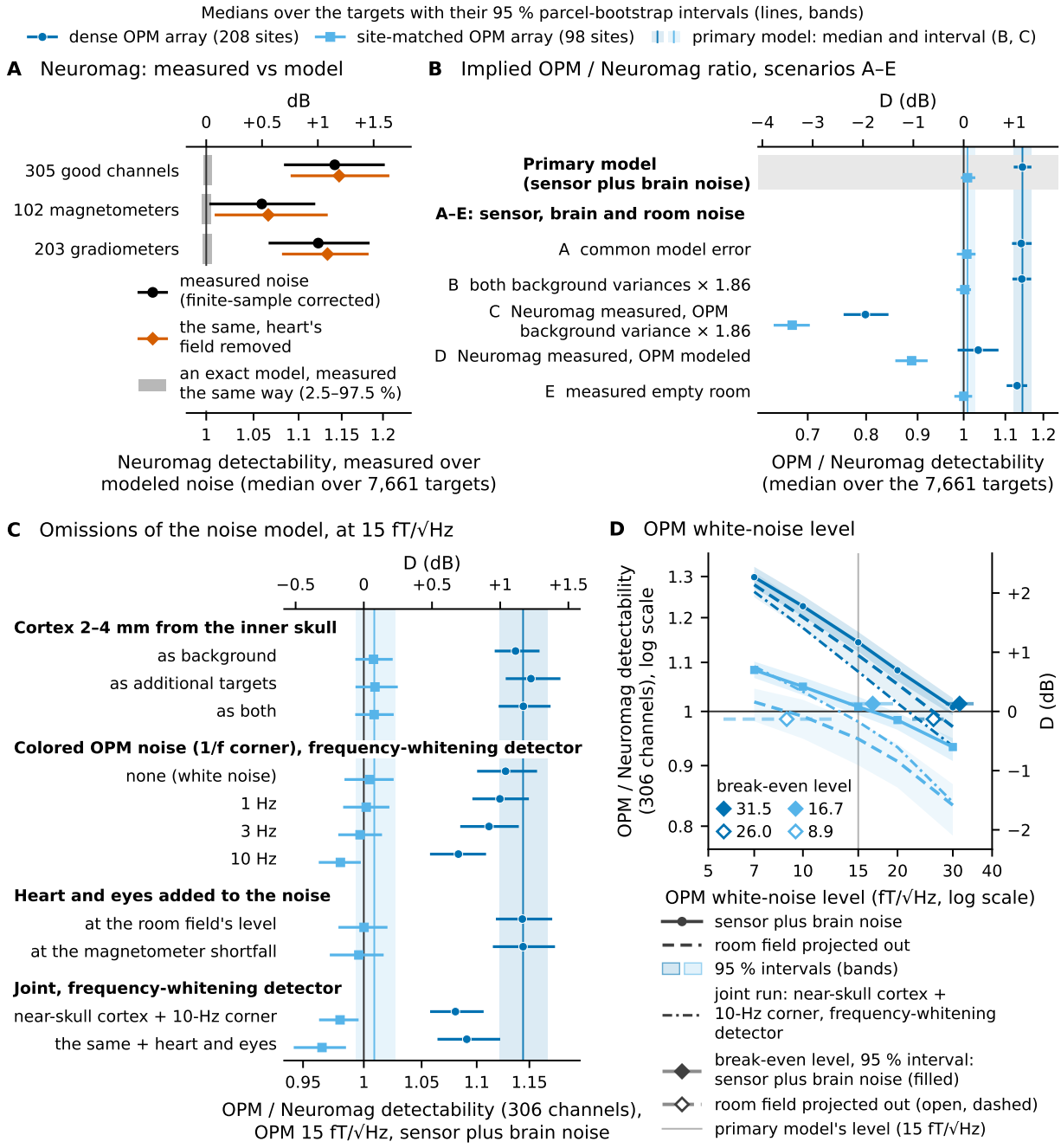


Fig. 4. The noise model against measured Neuromag noise. Adult at its measured head position; medians over the 7,661 targets (plus 463 near-skull targets in two rows of C) with 95 % parcel-bootstrap intervals; logarithmic ratio axes, with dB on the top or right axis. (A) Neuromag's detectability with its measured noise covariance over that with the modeled one (sensor, brain and room noise): 305 good channels, magnetometers, gradiometers. (B) The implied OPM/Neuromag ratio under scenarios A–E of Section 2.4 (305 channels; $k = 1.86$ scales a cortical background's variance; C is the least favorable to the OPM); gray row, and vertical lines and bands in B and C, the primary model (306 channels). (C) The ratio at 15 fT/√Hz with the model's omissions added: the near-skull cortex, 2–4 mm from the inner skull, as background, as additional targets or both; colored OPM noise and the joint runs under the frequency-whitening detector (flat signal spectrum); and cardiac and ocular sources together, at the room field's level at the median magnetometer or at the magnetometer brain-noise shortfall (Section S3.6). (D) The ratio against the OPM white-noise level; dash-dotted, the first joint run of C (no cardiac or ocular sources, no band). Break-even diamonds sit just above and below 1, keeping their intervals apart. Table 3 lists B and selected values of C.

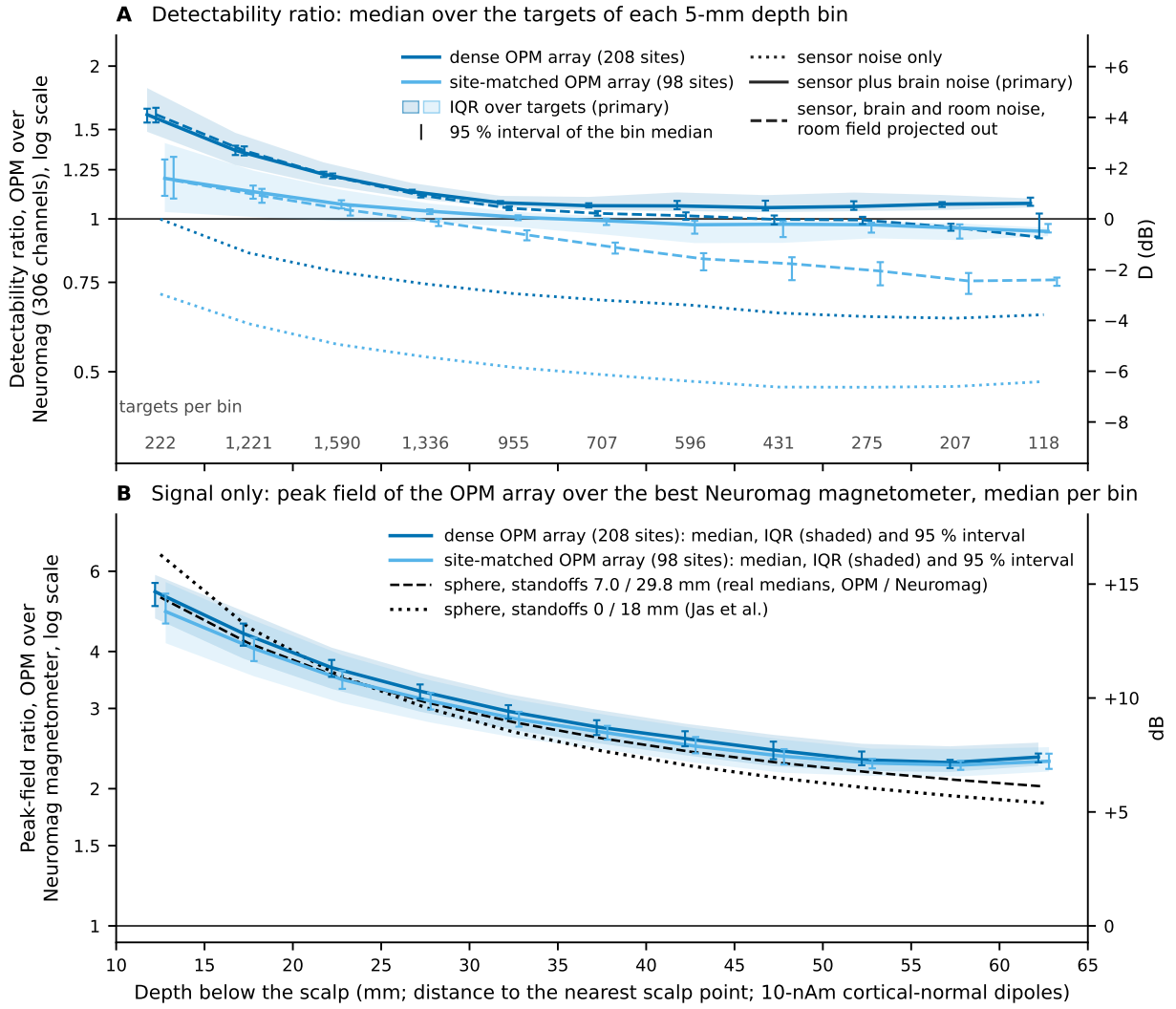


Fig. 5. The adult comparison by depth. (A) Median target-wise detectability ratio OPM/Neuromag (306 channels) in 5-mm depth bins for the dense (208 sites) and site-matched (98 sites) OPM arrays, with sensor noise only (dotted), sensor plus brain noise (solid; the primary condition, its interquartile range, IQR, over targets shaded) and the room field projected out (dashed). Bars, for the primary and projected conditions only: 95 % intervals of the bin median from 1,000 resamples of the parcel labels in the bin. Gray numbers: targets per bin (the few targets deeper than the last bin are not drawn). (B) Signal only: peak field of the OPM array over that of the best Neuromag magnetometer (median, IQR and 95 % interval per bin), with the analytical sphere (Section S2) at the real median standoffs (7.0 mm for the site-matched OPM sites, 29.8 mm for the Neuromag magnetometers; dashed) and at the standoffs of the spherical analysis (Jas et al., 2026) (dotted). Within each bin, the curves of the primary and projected conditions, with their bands and bars, are offset slightly along the depth axis so that the bars do not overlap (in B, those of the two arrays); the curves with sensor noise only stand at the bin centers; all use the same bins. Ratio axes are logarithmic, with dB on the right.

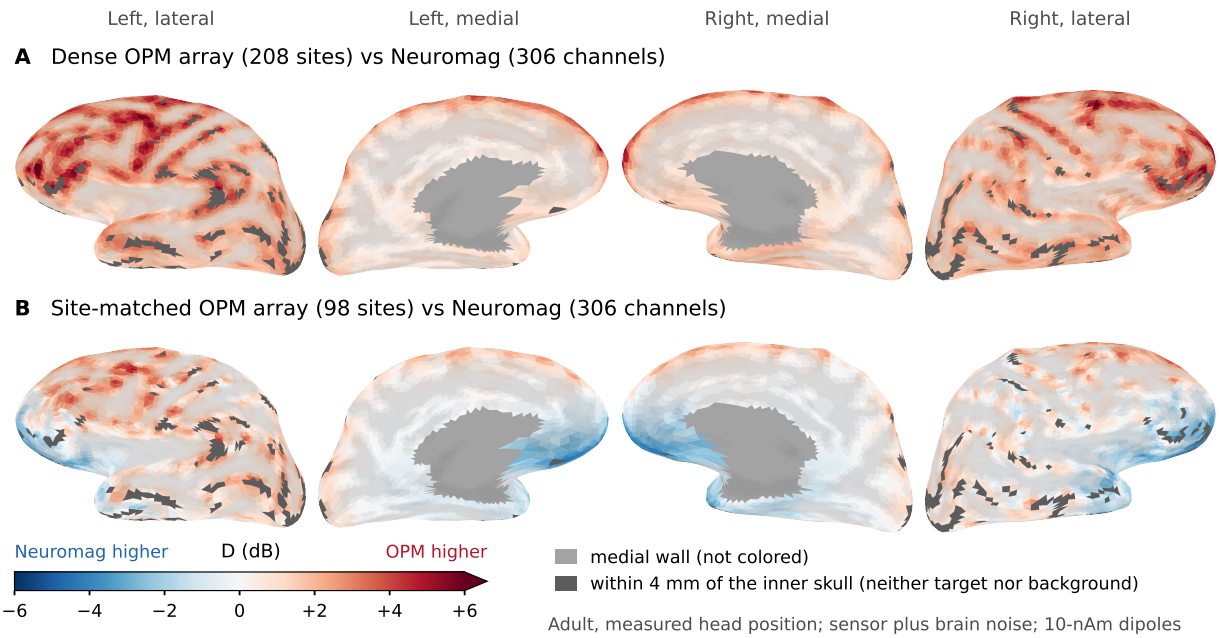


Fig. 6. The adult comparison on the cortex. *D* on the adult's inflated cortex at its measured head position, sensor plus brain noise, for (A) the dense and (B) the site-matched array against Neuromag's 306 channels; red, the OPM's detectability is higher. Median *D* is +1.17 dB (dense) and +0.08 dB (site-matched) over all 7,661 targets. Values above +6 dB take the end color (26 cortical targets of the dense array and 1 of the site-matched array). Each surface triangle is colored by the mean *D* of its targets; the shading of brightness shows the cortical folding and carries no data, and each view fills its panel (the views are not to one scale). Light gray, medial wall (its 558 targets enter the medians but are not colored); dark gray, cortex within 4 mm of the inner skull, neither a target nor part of the background.

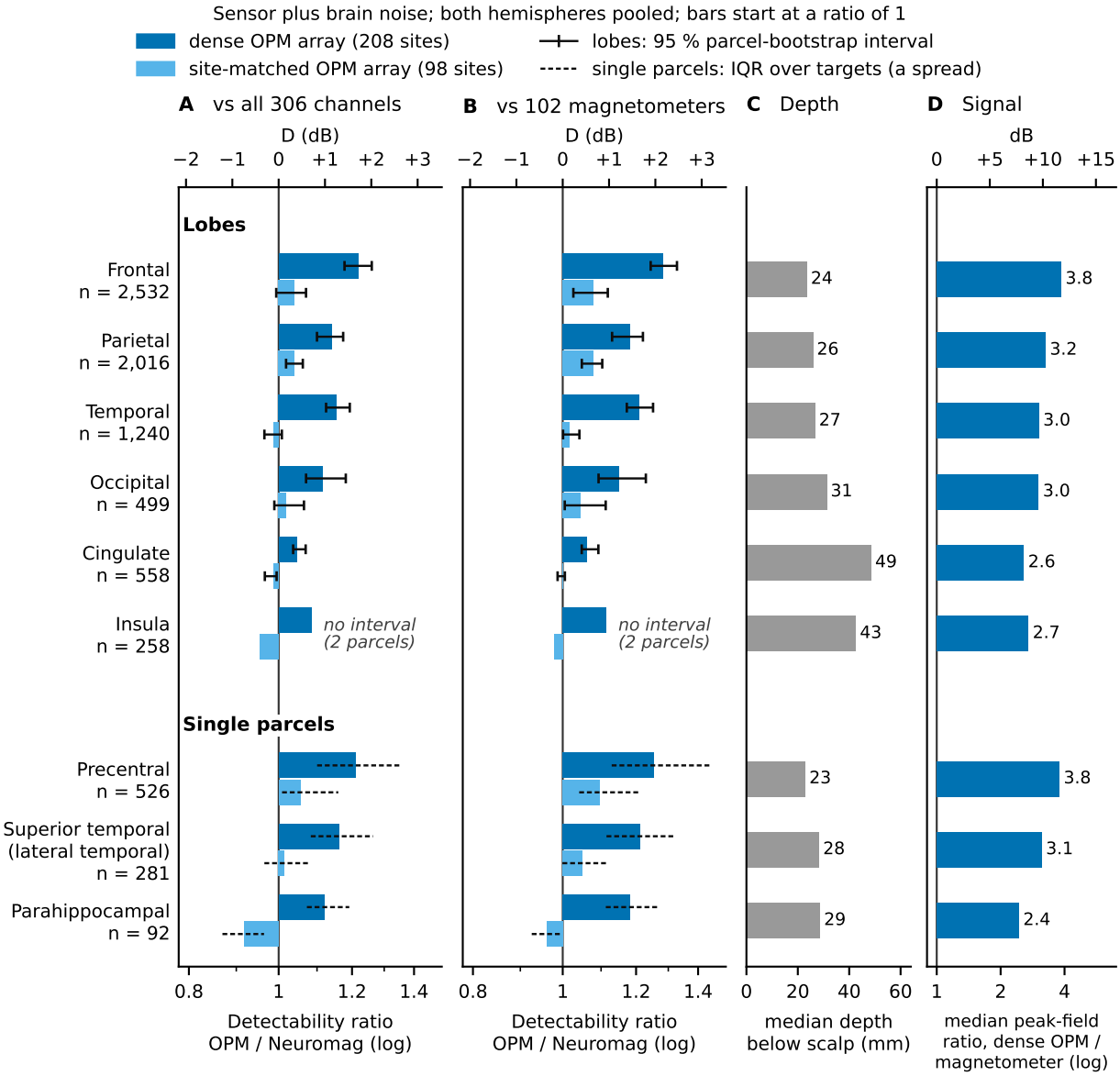


Fig. 7. The adult comparison by brain region. Adult at its measured head position; sensor plus brain noise, OPM noise of 15 fT/√Hz; hemispheres pooled. (A) Median detectability ratio OPM/Neuromag (306 channels) for the dense and site-matched arrays; bars start at 1 on a logarithmic axis, with D (dB) on the top axis. Lobes: median with 95 % parcel-bootstrap intervals (solid; the insula, with two parcels, has none). Single parcels (precentral; superior temporal, labeled lateral temporal; parahippocampal): median with the interquartile range over the parcel's targets (dashed), a spread rather than a confidence interval. (B) The same against the 102 magnetometers alone. (C) Median depth below the scalp (gray bars). (D) Median peak-field ratio of the dense array over the best magnetometer (signal only; blue bars from 1, logarithmic axis, with dB on the top axis). Targets per region: 2,532 frontal, 2,016 parietal, 1,240 temporal, 499 occipital, 558 cingulate, 258 insula, 526 precentral, 281 superior temporal and 92 parahippocampal.

Table 4. The principal smaller heads against the adult.

Head	Gap (mm)	D, fixed helmet (dB)	Δ, fixed helmet (dB)	Interaction (dB)	Δ, fitted helmet (dB)	Δ, fitted helmet, equal OPM sites (dB)
Adult	28.4	+1.00 [+0.82, +1.17]	–	–	–	–
School-age size	34.1	+1.50 [+1.28, +1.75]	+0.44 [+0.34, +0.56]	+0.40 [+0.30, +0.50]	+0.03 [0.00, +0.05]	+0.20 [+0.16, +0.24]
2-year size	39.1	+2.13 [+1.83, +2.44]	+1.07 [+0.91, +1.18]	+1.10 [+0.97, +1.21]	+0.02 [−0.02, +0.06]	+0.30 [+0.24, +0.35]
24-month template	38.5	+1.84 [+1.52, +2.19]	+0.73 [+0.46, +0.98]	+0.54 [+0.44, +0.73]	+0.04 [−0.06, +0.19]	+0.42 [+0.29, +0.68]
12-month template	40.4	+2.04 [+1.64, +2.43]	+0.96 [+0.61, +1.51]	+0.66 [+0.56, +0.99]	+0.05 [−0.09, +0.24]	+0.49 [+0.33, +0.81]

Dense OPM array against Neuromag (306 channels), sensor plus brain noise. Gap: median magnetometer-to-scalp gap in the fixed adult helmet at top contact. D: area-weighted median over cortical targets at top contact. Δ: paired change in D from the adult placed in the same helmet and pose (Section 2.1). Interaction: the Δ estimator applied to the within-head contrast F between the fixed and the fitted helmet (Eq. (3)), a paired median, not $\Delta_{\text{fixed}} - \Delta_{\text{fitted}}$. Fitted helmet: the helmet fitted at the gap of the adult; equal OPM sites: the dense array of the adult subsampled to the number of sites of each head. Brackets: 95 % parcel-bootstrap intervals; in the school-age size, the lower bound of the fitted-helmet Δ rounds to zero from above. The heads of Section S10 are listed in Tables S5–S7.

+0.48 to +1.07 dB, respectively (differences of medians, without intervals). These heads also had larger median magnetometer-to-scalp gaps: 34.1–40.4 mm, compared with 28.4 mm in the adult (Fig. 9A). Their larger OPM advantage was thus accompanied by a larger gap, which the next section separates from head size.

Head placement in the helmet, an uncertainty that the parcel-bootstrap intervals do not include, mattered as much as head size. Across the fixed helmet's source-blind placements, the school-age scaled adult's change from the adult spanned +0.20 to +1.12 dB (Fig. 9B). This range of differences of medians is wider than the parcel interval of the change at top contact, [+0.34, +0.56] dB. The change remained positive at every placement in each of the four principal heads (Fig. 9B, gray ranges). Across the same placements, the adult's own D spanned 0.46 dB. The four heads reported separately in Section S10 had a fixed-helmet Δ of +0.17 to +0.88 dB, with the smallest changes in the three school-aged children (+0.17 to +0.41 dB). A larger OPM advantage in a smaller head may therefore reflect adult-helmet fit as much as the head itself.

3.5. Helmet fit versus head size

In every principal smaller head, the fixed adult helmet was less favorable to Neuromag relative to the fitted helmet than it was in the adult (Fig. 9C; Table 4). The interaction ranged from +0.40 to +1.10 dB, with every interval above zero. The scaled adults defined the extremes, with the templates intermediate at +0.54 to +0.66 dB. In the four heads reported separately in Section S10, the interaction was +0.04 to +0.71 dB; only child C's interval included zero (+0.04 [−0.11, +0.19] dB).

Part of the interaction comes from the adult, whose contrast between the helmets was negative (−0.23 dB). This negative contrast is consistent with the adult's larger median gap in the fitted helmet (29.6 mm) than at top contact (28.4 mm). The negative adult contrast is the main reason the interaction exceeded each principal head's within-head contrast (+0.17 to +0.76 dB). About half of the adult contrast, however, arises from constructing the fitted helmet itself. Between top contact and a helmet fitted at the adult's top-contact gap, which have the same median gap, the

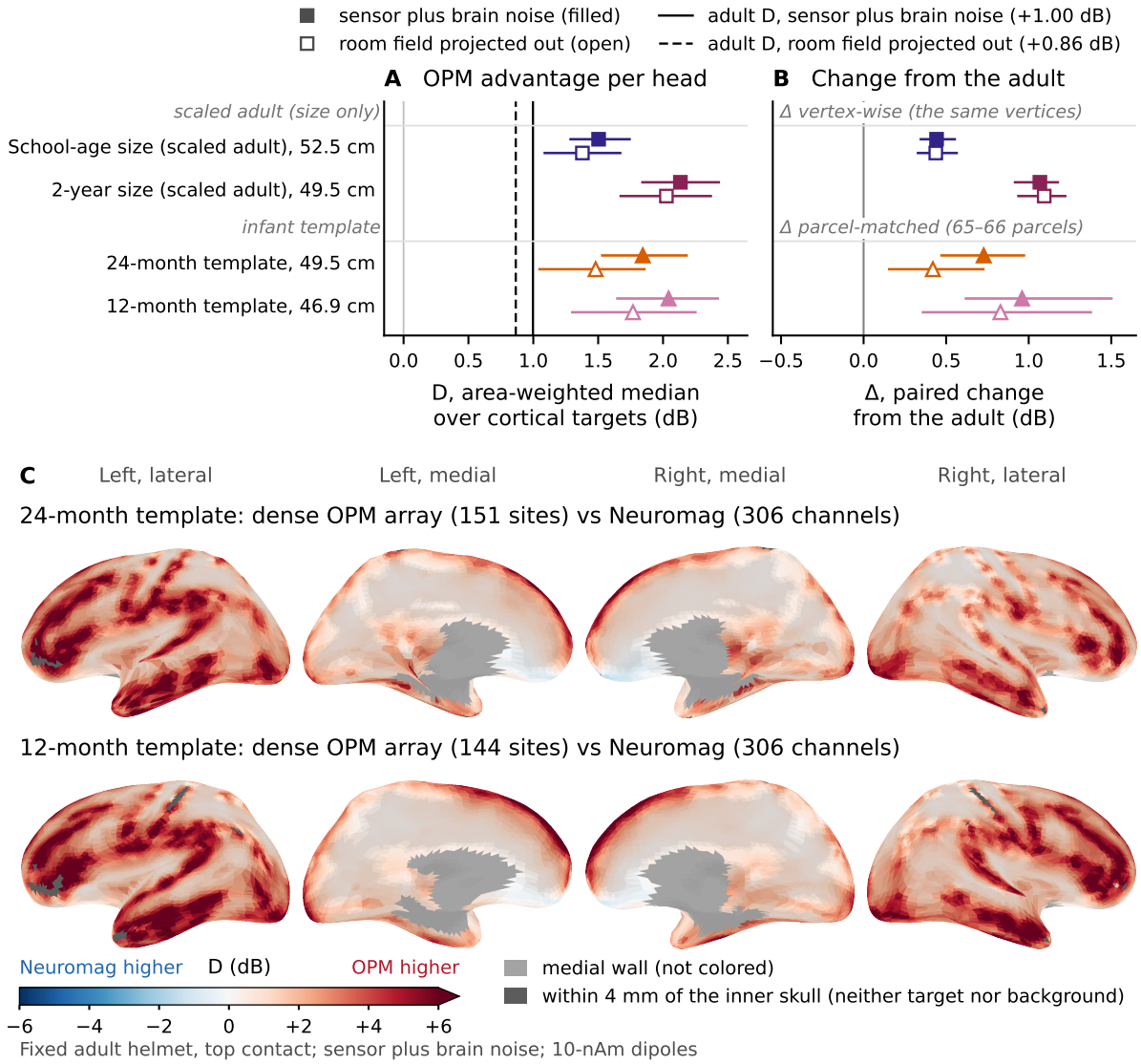


Fig. 8. Smaller heads in the fixed adult helmet. Dense OPM array against Neuromag (306 channels) at top contact. (A) *D* for the four principal smaller heads, the area-weighted median of the target-wise *D* over each head's cortical targets (medial wall excluded; target counts in Table 1), with sensor plus brain noise (filled) and with the room field projected out (open), with 95 % parcel-bootstrap intervals; vertical lines, the adult's *D* at top contact by the same convention (+1.00 and +0.86 dB). (B) Δ, the paired change from the adult (vertex-wise over the common vertices for the scaled adults; parcel-matched for the templates, over 66 and 65 parcels), in the same two conditions, with 95 % parcel-bootstrap intervals (1,000 resamples). Heads are grouped by class and ordered by head circumference (the number after each label); squares, scaled adults; triangles, infant templates; one color per head, used again in Fig. 10. (C) The target-wise *D* on the inflated cortex of the 24-month and 12-month infant templates (dense arrays of 151 and 144 sites), drawn as in Fig. 6; values above +6 dB take the end color (3.3 % and 6.1 % of the cortical targets). All nine anatomies are shown in Fig. S16. Infant templates: O'Reilly et al. (2021), from the Neurodevelopmental MRI Database (Richards et al., 2016).

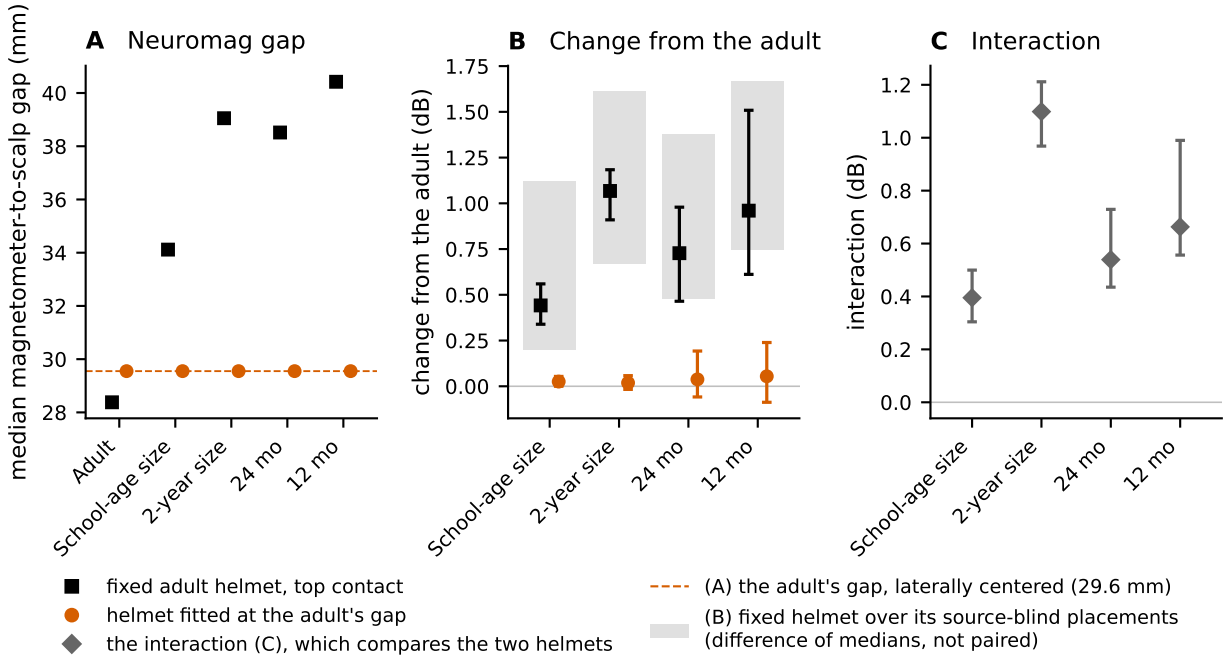


Fig. 9. Helmet fit versus head size. Dense OPM array against Neuromag (306 channels), sensor plus brain noise; bars, 95 % parcel-bootstrap intervals (for Δ and the interaction, of the paired differences). Filled black squares, the fixed adult helmet at top contact; filled vermilion circles, the helmet fitted at the adult's gap, built about the laterally centered pose (Section 2.3). (A) Median magnetometer-to-scalp gap in each helmet; dashed line, the adult's gap when laterally centered (29.6 mm). (B) Δ, the paired change from the adult in the same helmet and pose; gray ranges, the change from the adult over the 12 source-blind placements of the fixed helmet, computed as a difference of medians rather than as the paired Δ, and so not exactly comparable with the markers. (C) The interaction, the primary helmet-fit quantity (gray diamonds, because it compares the two helmets): the Δ estimator applied to the within-head contrast F of Eq. (3). Head labels: School-age size and 2-year size, the scaled adults; 24 and 12 mo, the infant templates. All nine anatomies are shown in Fig. S17.

contrast was −0.11 [−0.15, −0.08] dB (Section S5).

With each head in a fitted helmet and with its own dense array, most of the smaller heads' extra OPM advantage disappeared (Fig. 9B; Table 4). When both the smaller head's helmet and the adult's were fitted at the adult's gap, the change Δ was +0.02 to +0.05 dB, with interval bounds of −0.09 to +0.24 dB (Table 4). Every interval included zero except the school-age scaled adult's, whose lower bound was just above zero. Fitting instead at the adult's top-contact gap left a change of +0.10 to +0.23 dB relative to the adult at top contact (Section S5). With the room field projected out, the fitted-helmet change was negative in every principal head (−0.25 to −0.05 dB); only one of the four intervals included zero (Table S6). Thus, the fitted-helmet change from the adult was a few hundredths of a decibel and became negative after projecting out the room field.

The fitted helmet changes the sensor gap, placement and coverage together, so none of the quantities reported here apportions the effects of these factors exactly. At equal OPM site count, the fitted-helmet change increased to +0.20 to +0.49 dB (Table 4), compared with +0.02 to +0.05 dB with each head's own array and +0.44 to +1.07 dB in the fixed helmet. Fitting therefore removed almost all the extra advantage only because each smaller head had its own, smaller dense array. The comparison suggests that, in the fitted helmet, the scaled adults and templates retained a gain of a few tenths of a decibel from their smaller size, a gain that their fewer OPM sites offset.

Table 5. Detection of simulated superficial spikes.

Analysis	Adult	School-age size	2-year size	24-month template	12-month template
S50 ratio, Neuromag over the OPM array, by detector and thresholds					
Practical, frozen (endpoint)	1.45	1.44	1.47	1.28	1.39
	[1.21, 1.68]	[1.26, 1.58]	[1.24, 1.62]	[1.18, 1.45]	[1.25, 1.56]
five replicates pooled	1.41	1.44	1.42	1.25	1.31
	[1.22, 1.56]	[1.32, 1.53]	[1.25, 1.53]	[1.18, 1.37]	[1.22, 1.46]
Practical, re-fitted	1.47	1.50	1.42	1.27	1.42
	[1.22, 1.68]	[1.32, 1.63]	[1.22, 1.61]	[1.17, 1.42]	[1.27, 1.62]
Oracle	1.39	1.61	1.62	1.51	1.48
	[1.22, 1.55]	[1.46, 1.76]	[1.40, 1.78]	[1.31, 1.84]	[1.30, 1.84]
Mismatched, frozen	1.50	1.39	1.49	1.42	1.43
	[1.30, 1.67]	[1.23, 1.52]	[1.26, 1.67]	[1.26, 1.68]	[1.25, 1.60]
Mismatched, re-fitted	1.48	1.42	1.51	1.39	1.43
	[1.27, 1.64]	[1.26, 1.54]	[1.28, 1.69]	[1.24, 1.65]	[1.26, 1.63]
Site-matched array[a]	1.30	1.28	1.16	1.06	1.30
	[1.10, 1.46]	[1.10, 1.43]	[1.05, 1.35]	[0.95, 1.17]	[1.15, 1.48]
Exploratory run[b]	1.36	1.60	1.65	1.44	1.46
	[1.10, 1.58]	[1.31, 1.87]	[1.41, 1.87]	[1.20, 1.65]	[1.21, 1.84]
False events per minute, dense array / Neuromag					
Frozen thresholds	1.20 / 1.05	1.25 / 1.35	1.40 / 1.45	1.25 / 2.25	1.00 / 1.05
Re-fitted thresholds	1.20 / 0.95	1.60 / 0.70	1.00 / 1.25	1.15 / 1.45	1.40 / 0.90

Focal spikes 10–20 mm below the scalp; dense OPM array against Neuromag (306 channels) unless stated; above 1, the OPM detects at a lower strength. Pre-specified run: 36 locations per anatomy, first noise replicate unless stated (the pooled row uses all five replicates of the same locations). Thresholds of the practical and mismatched detectors were frozen for, or re-fitted on held-out null data to, a nominal 1 false event per minute, and the oracle detector tested at a per-trial false-positive probability of 0.001. False events per minute were evaluated on separate null data. The adult was at its measured head position, and the smaller heads at top contact in the fixed adult helmet. Brackets: 95 % location-bootstrap intervals. Holm-adjusted p values and location counts are listed in Table S11. [a]Practical detector, frozen thresholds. [b]Practical detector, frozen thresholds; 18 locations per anatomy, drawn independently of the pre-specified run, with one noise realization per event; endpoint chosen after the analyses (Section 2.6).

3.6. Simulated interictal spikes

Under the modeled noise, idealized detectors at a nominal false-event rate detected simulated superficial spikes at lower strength with the dense array than with Neuromag in the adult and every principal smaller head (Fig. 10A; Table 5). The pre-specified run's declared endpoint, the S50 ratio Neuromag/OPM, was 1.28–1.47 in these anatomies, with every location-bootstrap interval above 1 (Table 5). For spikes 10–20 mm deep, the dense array thus detected 50 % of events at a 22–32 % lower dipole strength. The smaller heads were at top contact in the fixed adult helmet, whose fit raises their modeled OPM advantage (Section 3.5). Their detection advantage therefore includes an effect of helmet fit; spikes were not simulated in the fitted helmet. As expected with higher detectability, the declared sign-flip test favored the dense array in each of these anatomies ($p < 0.0001$ in each of these five, Holm-adjusted over the nine declared anatomies; Table S11). In the adult, the S50 ratio (1.45) was of the order of the detectability ratio at these depths (1.36–1.60; Fig. 5A), so the detectability advantage carried over to the detection of these spikes.

The S50 ratio changed little across noise replicates, with the oracle and mismatched detectors, and with re-fitted thresholds (Fig. 10B, C; Table 5). Across five noise replicates, its standard deviation was 2.9–5.4 %, with pooled ratios of 1.25–1.44 (Table 5). The oracle detector favored the dense array by a similar or larger factor (1.39–1.62; Fig. 10B). Its threshold does not depend on the threshold calibration of the practical detector, and its realized false-positive probability

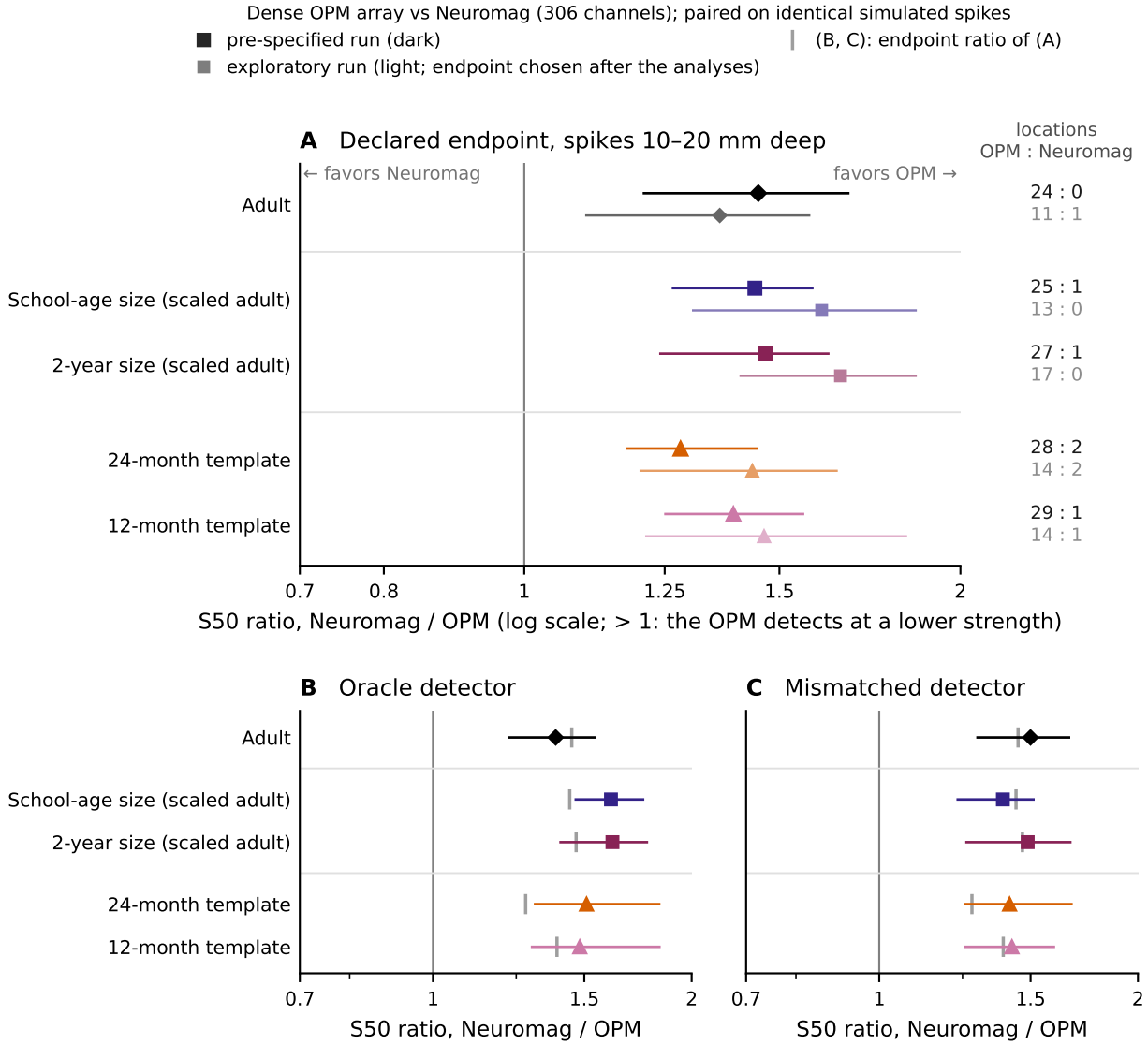


Fig. 10. Simulated interictal spikes: the pre-specified run. S50 ratio (S50, the dipole strength detected in 50 % of events), Neuromag (306 channels) over the dense OPM array, for focal spikes 10–20 mm below the scalp at 36 newly drawn locations per anatomy (648 focal events; the first of five noise replicates); logarithmic axis (above 1, the OPM detects at a lower strength); lines, 95 % location-bootstrap intervals (1,000 resamples). (A) The declared endpoint: the practical scanning detector, idealized in that it uses the true forward model and the simulated waveform family, with thresholds frozen for a nominal 1 false event per minute (dark markers). Right: the numbers of locations at which the OPM or Neuromag detected more events (the others are ties); the Holm-adjusted p values are in Table S11. Light markers: the exploratory run (18 locations per anatomy; endpoint chosen after the analyses). (B) The oracle detector (known source, waveform and time; per-trial false-positive probability 0.001). (C) The mismatched detector (Section 2.5), with frozen thresholds. In B and C, gray bars mark the endpoint's ratio of A. Diamond, the adult at its measured head position; other heads at top contact in the fixed adult helmet, with symbols and colors as in Fig. 8. The values are listed in Table 5. All nine anatomies are shown in Fig. S18.

per trial was 0.00093–0.0011 on held-out null data. Re-fitting thresholds on held-out null data to the nominal 1 false event per minute changed the ratios little (1.27–1.50). The mismatched detector, which departs from the simulation in waveform, forward model and coregistration, gave S50 ratios of 1.39–1.50 (Fig. 10C). These differed from the endpoint by a factor of 0.97–1.11, with every interval of the change including 1 (Table S14). With this detector, the test favored the dense array in all five anatomies, with frozen and with re-fitted thresholds.

The practical detector compared arrays at a nominal rather than equal realized false-event rate (Table 5). On separate null data, frozen thresholds produced 1.00–1.40 false events per minute for the dense array and 1.05–2.25 for Neuromag. An approximate equal-rate test, which we left uncorrected because the arrays share background and room noise, had $p < 0.05$ in one of these anatomies, the 24-month template. Neuromag had the higher rate in this anatomy, a departure that would, if anything, understate the dense array's advantage in the declared endpoint. After re-fitting, rates were 1.00–1.60 for the dense array and 0.70–1.45 for Neuromag. The test then had $p < 0.05$ in one anatomy, the school-age scaled adult, where the dense array had the higher rate. In this anatomy, the re-fitted ratio would therefore, if anything, overstate the dense array's advantage in this secondary analysis (Section S7).

The site-matched array had smaller S50 ratios than the dense array (1.06–1.30; Table 5), consistent with its detectability advantage being limited to shallow sources (Section 3.3). The exploratory and pre-specified runs agreed: in the same depth band, the exploratory dense-array S50 ratios were 1.36–1.65 across 18 locations per anatomy (Fig. 10A, light markers). Section S7 reports the deeper bands of the exploratory run. In the pre-specified run, the four heads reported separately in Section S10 had ratios of 1.27–1.43. Thus, the dense array detected superficial spikes at lower strength than Neuromag with every detector, with frozen and re-fitted thresholds, and in both runs.

In contrast to detection, no localization difference between arrays survived correction over all comparisons in the exploratory run. That run covered all nine anatomies, including the four reported separately in Section S10. Dipole fits of detected 320-nAm focal spikes had median errors of 3.7–7.6 mm in every array and anatomy, of the order of the coregistration displacement (2.1–3.2 mm). Of 360 paired comparisons of dipole and dSPM errors and of joint detection and localization, 65 reached $p < 0.05$ uncorrected, compared with about 18 expected by chance. Most nominally significant comparisons favored an OPM array (61). Only eight, however, survived Bonferroni correction within anatomy and array, and none survived correction over all 360 (Section S7; Fig. S14).

4. Discussion

In our simulations of a realistic adult head, the comparison between on-scalp OPMs and Neuromag depended on the array design, the OPM noise floor, the Neuromag channels used as comparator and the SNR measure. Brain noise turned the SQUID advantage into an advantage of the dense OPM array, but the dense-array advantage was as large as the model's error for Neuromag's noise. When helmets were fitted at the adult's gap and each head had its own dense array, most of the smaller heads' extra OPM advantage disappeared. Under the modeled noise, the detectability difference carried over to the detection of superficial simulated spikes. Together, these results suggest that proximity alone does not set the on-scalp advantage: array design, sensor noise and SQUID helmet fit determine whether it appears and how large it is.

4.1. What determines the adult comparison

The spherical analysis of Jas et al. (2026), reimplemented in Section S2 (Fig. S1), places the OPM advantage within an equal-SNR depth set by an assumed noise ratio. Under modeled sensor

and brain noise, however, the dense array had an advantage at every depth. With the room field projected out, the dense array had an equal-SNR depth, from the 45–50 mm bin; with Neuromag's measured noise, it had one in point estimates, from the 25–30 mm bin. Whether the dense array's advantage extends to deep sources therefore depends on the noise model, whereas the site-matched array had an equal-SNR depth even with sensor plus brain noise. The OPM noise floor and the array design, in contrast, can reverse which system has the higher detectability over the whole cortex.

The reversal by brain noise is consistent with each channel's noise budget. Relative to their sensor floors, brain noise raises Neuromag magnetometer noise far more than OPM noise (Section 3.1), making the OPM's higher sensor noise less consequential once brain noise dominates. The comparator and SNR measure then determine which system a reported SNR favors. Peak-channel SNR favors OPMs when the comparator uses magnetometers and Neuromag when it uses the best of all channels; channel-averaged SNR shows little difference.

The size of the dense array's advantage cannot be determined more precisely than the model's error for Neuromag noise allows. The five scenarios describe assumptions about how this error is shared, rather than bounds on it, and we cannot determine which comes closest here. Scenario D, with measured Neuromag noise and modeled OPM noise, is not the case least favorable to OPMs: OPMs are magnetometers, and their real noise may share some of the far-field excess that the model misses for Neuromag's magnetometers.

The break-even levels inherit this dependence on the noise model, because they were computed for single-axis arrays with white OPM noise and modeled Neuromag noise. If Neuromag's measured noise applies, the dense array has no clear advantage at the primary OPM noise. At the point estimate, its break-even level would then lie above that noise by an amount the medians cannot determine, whereas the site-matched level would lie below it, if within the sweep at all (Section 3.2). To place a real array relative to Neuromag, one would therefore need to enter its measured noise covariance into the full model, with matched geometry, sensing axes and preprocessing. A colored and correlated covariance cannot be reduced to the sweep's single white-noise level.

4.2. Head size and helmet fit

The spherical analysis predicts a larger OPM benefit in smaller heads because, at a constant SQUID-to-scalp distance, more of a small brain lies within the equal-SNR depth (Jas et al., 2026). In the fixed adult helmet, the scaled adults and the two templates did show a larger OPM advantage than the adult. When the helmet was fitted to each head at the adult's gap and each head had its own dense array, however, only a few hundredths of a decibel of this excess remained. Relative to the fitted helmet, the fixed helmet was less favorable to Neuromag in every principal head than in the adult (Section 3.5). This comparison uses two Neuromag helmets around the same OPM array, so it is independent of unmeasured OPM noise, although dependent on modeled Neuromag noise and anatomy. It thus suggests that, in this model, the sphere's prediction holds in the adult helmet but largely reflects how that helmet fits a smaller head, through its gap, placement and coverage together. The construction neither isolates median gap nor excludes head size as a cause.

The scaled adults provide the main support for our conclusion because they preserve the adult anatomy (Table 4). The fitted-helmet residual depends on the adult's reference placement (Section 3.5) and, for templates, on scalp convention (Section 4.4). This small residual masks a gain from head size itself. At equal OPM site count, the fitted-helmet change was a few tenths of a decibel, up to about half of the fixed-helmet excess in the templates. With brain noise included, this is the analog of the head-size gain Zahran et al. (2022) found with sensor noise alone in a constant-gap helmet. Fewer OPM sites offset this gain in the smaller heads, and the gain is smaller than the interaction in each principal head.

We found this gain in uniformly scaled heads and smooth infant templates, whose cortex shrinks with the head. The school-aged children of Section S10 had head circumferences similar to those of the scaled heads and templates but nearly adult cortical areas (1,666–1,869 cm^2). At equal OPM site count, children A and B retained smaller gains (+0.16 and +0.16 dB, with intervals including zero). Their fewer OPM sites outweighed these gains, leaving negative fitted-helmet changes with their own arrays (−0.30 and −0.39 dB). Their near-scalp anatomy could not be verified, but they suggest that the gain may be smaller in individual children than in scaled heads and templates.

For pediatric MEG, these results suggest that much of the extra on-scalp advantage of small heads in an adult helmet reflects helmet fit. Our fitted helmet retains the adult's sensors and does not model a pediatric SQUID system, such as one built for young children (Tesan et al., 2010). A child's position in an adult helmet also matters (Gaetz et al., 2008). Three of the four separately reported heads also had a clearly positive interaction, whereas child C did not; the three school-aged children gained less in the fixed helmet than the principal heads (Section S10).

4.3. Simulated spikes and the clinical literature

Under the modeled noise and with idealized detectors, the dense array detected superficial spikes at lower strength than Neuromag in the adult and every principal smaller head. We compared the S50 ratio with the detectability ratio only in the adult, in which the S50 ratio was of the order of the detectability ratio at these depths. The ranking persisted in the adult and all principal smaller heads with the mismatched and oracle detectors and with re-fitted thresholds. The simulations thus illustrate that the modeled detectability difference carries over to detection at a nominal false-event rate. They neither test the noise model nor show how detection would fare under the noise alternatives of Section 3.2. In contrast to detection, no localization comparison survived correction over all such comparisons, although nominally significant comparisons were more frequent than expected by chance and almost all favored an OPM array. Dipole-fit errors were of the order of the coregistration error, which may have dominated the comparison.

We did not model clinical dual- or triaxial arrays, so the model does not determine where they would fall. The single-axis site-matched array had no whole-cortex advantage at the primary noise level and a disadvantage at higher OPM noise. Small advantages of this array appeared only at lower OPM noise, with some comparators and SNR measures, with a plug-in covariance, or within about 30 mm of the scalp. With three equally quiet axes at the same sites (294 channels), its ratio approached that of the dense array (Section 3.1). The triaxial result suggests that the number of independent measurements, from more sites or more axes per site, mattered more than site count alone, provided the extra axes are as quiet as the normal one. We obtained the dense array's modest whole-cortex advantage with more sites than any array used in the epilepsy studies of Table S19.

The clinical reports and our model are compatible, but the model cannot identify which factor operated in any study. Table S19 pairs each study with its closest model analog. Higher or similar OPM SNR against SQUID magnetometers (Feys et al., 2022; mixed results in Feys et al., 2025) corresponds to modeled peak-channel SNR against magnetometers. This SNR favors the dense array in the adult, every principal smaller head and the heads of Section S10 (Section S6). In contrast, equal or lower SNR against planar gradiometers or the nearest channels of few or noisier sensors (Hillebrand et al., 2023; Schwartz et al., 2025) corresponds to peak-channel SNR against the gradiometers or the best channel. The larger gains reported in school-aged children recorded in an adult helmet (Feys et al., 2022) are consistent with the fixed-helmet effect in our scaled adults and templates. However, the three school-aged children we modeled showed the smallest such effect (Section S10).

The weaker OPM results for temporal and mesial temporal sources (Feys et al., 2024, 2025; Shen et al., 2026) have two analogs in our model: the weakness of deep sources for every system and the site-matched array's deficit on ventral and polar surfaces. The latter is fragile because the model overstates Neuromag's detectability in the temporal lobe (Section 3.2). More generally, the regional ranking is exploratory and changes with the noise model. With Neuromag's measured noise (scenario D), the temporal lobe had the highest dense-array ratio, while cingulate and parietal ratios fell below 1 (Section 3.3). No regional recommendation therefore follows from this ranking.

4.4. Limitations

Several limitations remain, most of which point to a specific measurement or simulation. The most important concerns the noise model: no measurement validates the OPM covariance, OPM-specific noise (cross-talk, gain and sensitive-axis errors) was not modeled, and the model was calibrated and checked on one recording whose magnetometer noise it partly misses. Because the dense array's advantage is as large as the model's error for Neuromag, these limitations could reverse the adult comparison, whereas the omitted OPM noise does not enter the interaction. Measured OPM covariance and recordings from further subjects would test the advantage and the generality of the error. We also used one projection in place of real-system preprocessing (Taulu and Simola, 2006; Tierney et al., 2021), and clinical dual- and triaxial arrays were not modeled. Turning to anatomy, the pediatric conclusion rests on one adult, two uniformly scaled copies and two templates (Section 4.2); a cohort of individual pediatric MRIs would test it. The templates' scalp convention differs from the adult's by 2.6–3.0 mm (Section 2.2). This could bias their D, Δ and fitted-helmet residual (+0.04 and +0.05 dB) by an unquantified amount; its effect on the interaction, expected to be smaller, was not tested. The smaller heads keep the adult's background variance per unit area (Section 3.4), and all sources lie on the cortical surface. In every head, the primary model omits cortex within 4 mm of the inner skull, including the gyral crowns nearest the sensors; only adult sensitivity analyses restored it from 2 mm. Finally, the spike detectors use the simulation's stationary model, whose null data contain no physiological transients, and the threshold-calibration uncertainty is not propagated into the S50 intervals. Spikes were simulated neither under the noise alternatives nor in the fitted helmet, which recordings of real or phantom spikes with both systems would address.

5. Conclusions

We compared on-scalp OPM arrays with a cryogenic Neuromag system in simulations with explicit sensor, brain and room noise. In the adult, the array design and the OPM noise floor determined which system had the higher detectability. The comparator and the SNR measure determined which system a reported SNR favored. The dense single-axis array's modeled advantage was as large as the model's error for Neuromag noise. Its break-even OPM noise of 31.5 fT/$\sqrt{\text{Hz}}$ therefore applies only under modeled Neuromag noise; with measured Neuromag noise, the dense array had no clear advantage at 15 fT/$\sqrt{\text{Hz}}$. The single-axis site-matched array showed no clear difference from Neuromag, whereas three equally quiet axes at those sites approached the dense array; clinical dual- and triaxial arrays were not modeled. With each smaller head's own dense array, fitting the SQUID helmet removed most of the extra OPM advantage produced by the adult helmet. Under the modeled noise, idealized detectors detected weaker superficial spikes with the dense array than with Neuromag at a nominal false-event rate. Measured OPM covariances would test the adult comparison, and comparisons in children depend on how well the SQUID helmet fits.

CRediT authorship contribution statement

Laiba Khan: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Mainak Jas:** Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

The analysis code, configurations and tests are available at https://github.com/khan-laiba/opm-squid-meg under the MIT license, and the result files (JSON and CSV, in the repository's results folder) and figures under the CC BY 4.0 license; the result files reported here are those of commit b85dcdf. The external data are not redistributed but are obtained from their sources with the repository's fetch scripts: the MNE sample dataset (Gramfort et al., 2013), in the processed release that the dataset fetcher of MNE-Python 1.13.2 downloads (OSF file 86qa2, version 6); the infant templates (O'Reilly et al., 2021; Richards et al., 2016), through MNE-Python's template fetcher (LGPL-2.1); and the MRIs and surfaces of the children, OpenNeuro ds005234, version 2.2.0 (Fadeev et al., 2025), whose metadata give CC0 and whose authors ask that it be acknowledged as distributed under CC BY with its DOI. The documentation of the sample dataset restricts it to learning the software and advises against using it to evaluate the MEG or MRI system. Here, the recording calibrates the noise model and its measured covariance serves as a check of that model, and the values are not offered as an evaluation of the VectorView system or of the MRI scanner.

Ethics statement

This simulation study used only publicly available, de-identified data collected under the approvals of the original studies; no new data were collected from human participants, so no further ethical approval was required. Sex and gender were not analyzed: with one adult, three averaged templates and three children, the head models are too few for any analysis by sex.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to write and run the analysis code and to draft and edit the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

Alem, O., Hughes, K.J., Buard, I., Cheung, T.P., Maydew, T., Griesshammer, A., Holloway, K., Park, A., Lechuga, V., Coolidge, C., Gerginov, M., Quigg, E., Seames, A., Kronberg, E., Teale, P., Knappe, S., 2023. An integrated full-head OPM-MEG system based on 128 zero-field sensors. Front. Neurosci. 17, 1190310. https://doi.org/10.3389/fnins.2023.1190310.

Boto, E., Meyer, S.S., Shah, V., Alem, O., Knappe, S., Kruger, P., Fromhold, T.M., Lim, M., Glover, P.M., Morris, P.G., Bowtell, R., Barnes, G.R., Brookes, M.J., 2017. A new generation of magnetoencephalography: room temperature measurements using optically-pumped magnetometers. NeuroImage 149, 404–414. https://doi.org/10.1016/j.neuroimage.2017.01.034.

Dale, A.M., Liu, A.K., Fischl, B.R., Buckner, R.L., Belliveau, J.W., Lewine, J.D., Halgren, E., 2000. Dynamic statistical parametric mapping: combining fMRI and MEG for high-resolution imaging of cortical activity. Neuron 26, 55–67. https://doi.org/10.1016/S0896-6273(00)81138-1.

Desikan, R.S., Ségonne, F., Fischl, B., Quinn, B.T., Dickerson, B.C., Blacker, D., Buckner, R.L., Dale, A.M., Maguire, R.P., Hyman, B.T., Albert, M.S., Killiany, R.J., 2006. An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral based regions of interest. NeuroImage 31, 968–980. https://doi.org/10.1016/j.neuroimage.2006.01.021.

Elekta Neuromag, n.d. Elekta Neuromag TRIUX. Datasheet NM23083B-A. Elekta Oy. Helsinki. https://natmeg.se/onewebmedia/NM23083B-A%20Elekta%20Neuromag%20TRIUX%20datasheet.pdf (accessed 4 October 2026).

Fadeev, K.A., Romero Reyes, I.V., Goiaeva, D.E., Obukhova, T.S., Ovsiannikova, T.M., Prokofyev, A.O., Rytikova, A.M., Novikov, A.Y., Kozunov, V.V., Stroganova, T.A., Orekhova, E.V., 2024. Attenuated processing of vowels in the left temporal cortex predicts speech-in-noise perception deficit in children with autism. J. Neurodev. Disord. 16, 67. https://doi.org/10.1186/s11689-024-09585-2.

Fadeev, K.A., Romero Reyes, I.V., Goiaeva, D.E., Obukhova, T.S., Ovsiannikova, T.M., Prokofyev, A.O., Rytikova, A.M., Novikov, A.Y., Kozunov, V.V., Stroganova, T.A., Orekhova, E.V., 2025. Perception of vowel sounds in children with autism spectrum disorders and typically developing children (MEG/ERF study). [dataset] OpenNeuro, ds005234, version 2.2.0. https://doi.org/10.18112/openneuro.ds005234.v2.2.0.

Feys, O., Corvilain, P., Aeby, A., Sculier, C., Holmes, N., Brookes, M., Goldman, S., Wens, V., De Tiège, X., 2022. On-scalp optically pumped magnetometers versus cryogenic magnetoencephalography for diagnostic evaluation of epilepsy in school-aged children. Radiology 304, 429–434. https://doi.org/10.1148/radiol.212453.

Feys, O., De Tiège, X., 2024. From cryogenic to on-scalp magnetoencephalography for the evaluation of paediatric epilepsy. Dev. Med. Child Neurol. 66, 298–306. https://doi.org/10.1111/dmcn.15689.

Feys, O., Ferez, M., Corvilain, P., Schuind, S., Rikir, E., Legros, B., Gaspard, N., Holmes, N., Brookes, M., Wens, V., De Tiège, X., 2024. On-scalp magnetoencephalography based on optically pumped magnetometers can detect mesial temporal lobe epileptiform discharges. Ann. Neurol. 95, 620–622. https://doi.org/10.1002/ana.26844.

Feys, O., Wens, V., Depondt, C., Rikir, E., Gaspard, N., Van Paesschen, W., Aeby, A., Bodart, O., Carrette, E., Holmes, N., Brookes, M., Ferez, M., Corvilain, P., De Tiège, X., 2025. On-scalp magnetoencephalography based on optically pumped magnetometers to investigate temporal lobe epilepsy. Epilepsia 66, e142–e151. https://doi.org/10.1111/epi.18439.

Fischl, B., 2012. FreeSurfer. NeuroImage 62, 774–781. https://doi.org/10.1016/j.neuroimage.2012.01.021.

Gaetz, W., Otsubo, H., Pang, E., 2008. Magnetoencephalography for clinical pediatrics: the effect of head positioning on measurement of somatosensory-evoked fields. Clin. Neurophysiol. 119, 1923–1933. https://doi.org/10.1016/j.clinph.2008.04.291.

Goldenholz, D.M., Ahlfors, S.P., Hämäläinen, M.S., Sharon, D., Ishitobi, M., Vaina, L.M., Stufflebeam, S.M., 2009. Mapping the signal-to-noise-ratios of cortical sources in magnetoencephalography and electroencephalography. Hum. Brain Mapp. 30, 1077–1086. https://doi.org/10.1002/hbm.20571.

Gramfort, A., Luessi, M., Larson, E., Engemann, D.A., Strohmeier, D., Brodbeck, C., Goj, R., Jas, M., Brooks, T., Parkkonen, L., Hämäläinen, M., 2013. MEG and EEG data analysis with MNE-Python. Front. Neurosci. 7, 267. https://doi.org/10.3389/fnins.2013.00267.

Hämäläinen, M.S., Sarvas, J., 1989. Realistic conductivity geometry model of the human head for interpretation of neuromagnetic data. IEEE Trans. Biomed. Eng. 36, 165–171. https://doi.org/10.1109/10.16463.

Harris, C.R., Millman, K.J., van der Walt, S.J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N.J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M.H., Brett, M., Haldane, A., del Río, J.F., Wiebe, M., Peterson, P., Gérard-Marchant, P., Sheppard, K., Reddy, T., Weckesser, W., Abbasi, H., Gohlke, C., Oliphant, T.E., 2020. Array programming with NumPy. Nature 585, 357–362. https://doi.org/10.1038/s41586-020-2649-2.

Hill, R.M., Boto, E., Holmes, N., Hartley, C., Seedat, Z.A., Leggett, J., Roberts, G., Shah, V., Tierney, T.M., Woolrich, M.W., Stagg, C.J., Barnes, G.R., Bowtell, R., Slater, R., Brookes, M.J., 2019. A tool for functional brain imaging with lifespan compliance. Nat. Commun. 10, 4785. https://doi.org/10.1038/s41467-019-12486-x.

Hillebrand, A., Holmes, N., Sijsma, N., O'Neill, G.C., Tierney, T.M., Liberton, N., Stam, A.H., van Klink, N., Stam, C.J., Bowtell, R., Brookes, M.J., Barnes, G.R., 2023. Non-invasive measurements of ictal and interictal epileptiform activity using optically pumped magnetometers. Sci. Rep. 13, 4623. https://doi.org/10.1038/s41598-023-31111-y.

Holm, S., 1979. A simple sequentially rejective multiple test procedure. Scand. J. Stat. 6, 65–70.

Hunold, A., Funke, M.E., Eichardt, R., Stenroos, M., Haueisen, J., 2016. EEG and MEG: sensitivity to epileptic spike activity as function of source orientation and depth. Physiol. Meas. 37, 1146–1162. https://doi.org/10.1088/0967-3334/37/7/1146.

Iivanainen, J., Stenroos, M., Parkkonen, L., 2017. Measuring MEG closer to the brain: performance of on-scalp sensor arrays. NeuroImage 147, 542–553. https://doi.org/10.1016/j.neuroimage.2016.12.048.

Jas, M., Matsubara, T., Sohrabpour, A., Sundaram, P., Mody, M., Ahlfors, S.P., 2026. Signal-to-noise ratio of event-related fields in on-scalp and off-scalp MEG. bioRxiv preprint 2026.08.17.744953. https://doi.org/10.64898/2026.08.17.744953.

Ledoit, O., Wolf, M., 2004. A well-conditioned estimator for large-dimensional covariance matrices. J. Multivar. Anal. 88, 365–411. https://doi.org/10.1016/S0047-259X(03)00096-4.

Matsubara, T., Sohrabpour, A., Ahlfors, S.P., Jas, M., Samuelsson, J., Sundaram, P., Stufflebeam, S., 2026. Quantifying cerebellar signal detectability in MEG and EEG in epilepsy using anatomically informed source modeling. NeuroImage 333, 121930. https://doi.org/10.1016/j.neuroimage.2026.121930.

de Munck, J.C., Vijn, P.C.M., Lopes da Silva, F.H., 1992. A random dipole model for spontaneous brain activity. IEEE Trans. Biomed. Eng. 39, 791–804. https://doi.org/10.1109/10.148387.

O'Reilly, C., Larson, E., Richards, J.E., Elsabbagh, M., 2021. Structural templates for imaging EEG cortical sources in infants. NeuroImage 227, 117682. https://doi.org/10.1016/j.neuroimage.2020.117682.

Osborne, J., Orton, J., Alem, O., Shah, V., 2018. Fully integrated, standalone zero field optically pumped magnetometer for biomagnetism. Proc. SPIE 10548, 105481G. https://doi.org/10.1117/12.2299197.

Ren, J., Ding, M., Peng, Y., Sun, C., Yang, C., Zhou, S., Tian, J., Wang, Q., Li, Z., 2025. A comparative study on the detection and localization of interictal epileptiform discharges in magnetoencephalography using optically pumped magnetometers versus superconducting quantum interference devices. NeuroImage 312, 121232. https://doi.org/10.1016/j.neuroimage.2025.121232.

Richards, J.E., Sanchez, C., Phillips-Meek, M., Xie, W., 2016. A database of age-appropriate average MRI templates. NeuroImage 124, 1254–1259. https://doi.org/10.1016/j.neuroimage.2015.04.055.

Schwartz, D., Badier, J.M., Gutteling, T.P., Daligault, S., Labyt, E., Bonini, F., Jung, J., 2025. Potential of ^{4}He OPM-MEG for detecting interictal epileptic activity: a comparison with SQUID-MEG. Epilepsia Open 10, 1660–1672. https://doi.org/10.1002/epi4.70139.

Shen, Y., You, C., Zhang, Y., Ji, N., Zhao, X., Zhang, P., Huang, S., Kang, H., Liu, X., Peng, Y., Sun, C., Yan, B., Zhang, Y., Zhu, S., Zhu, W., Lei, T., Tang, Z., Ding, M., Hu, F., Shu, K., 2026. Assessment of the utility of optically pumped magnetometer magnetoencephalography in preoperative localization of refractory epilepsy: a prospective study. Epilepsia 67, 3937–3954. https://doi.org/10.1002/epi.70273.

Taulu, S., Simola, J., 2006. Spatiotemporal signal space separation method for rejecting nearby interference in MEG measurements. Phys. Med. Biol. 51, 1759–1768. https://doi.org/10.1088/0031-9155/51/7/008.

Tesan, G., Johnson, B.W., Reid, M., Thornton, R., Crain, S., 2010. Measurement of neuromagnetic brain function in pre-school children with custom sized MEG. J. Vis. Exp. 36, 1693. https://doi.org/10.3791/1693.

Tierney, T.M., Alexander, N., Mellor, S., Holmes, N., Seymour, R., O'Neill, G.C., Maguire, E.A., Barnes, G.R., 2021. Modelling optically pumped magnetometer interference in MEG as a spatially homogeneous magnetic field. NeuroImage 244, 118484. https://doi.org/10.1016/j.neuroimage.2021.118484.

Westin, K., Beniczky, S., Pfeiffer, C., Hämäläinen, M., Lundqvist, D., 2023. On the clinical utility of on-scalp MEG: a modeling study of epileptic activity source estimation. Clin. Neurophysiol. 156, 143–155. https://doi.org/10.1016/j.clinph.2023.10.006.

Zahran, S., Mahmoudzadeh, M., Wallois, F., Betrouni, N., Derambure, P., Le Prado, M., Palacios-Laloy, A., Labyt, E., 2022. Performance analysis of optically pumped ^{4}He magnetometers vs. conventional SQUIDs: from adult to infant head models. Sensors 22, 3093. https://doi.org/10.3390/s22083093.

Appendix A. Supplementary material

Supplementary material related to this article (Sections S1–S10, Figs. S1–S18 and Tables S1–S19) is provided as a separate file.